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PROJECT FINAL REPORT

Name: Enuka Geenad Meemanage

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Network Solution

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Abstract

ABC Holdings PLC is hindered by multiple vulnerabilities to its network security and infrastructure. The vulnerabilities are caused by, among other things, a lack of VLAN separation in the current set, a lack of Wi-Fi security, and poor inter-departmental communication controls. The main server is also vulnerable with physical security far too weak and network security potential vulnerabilities could come from weaknesses in both standards. To improve security, compliance, and efficiency/operations as it is the company needs to do a near total redesign of their network. This proposal outlines plans for a set of VLANs to control department separation with access control, port security and a firewall to control access and phishing attacks, as well as policy management. Our recommendations also plan for an infrastructure that has a secure wireless setup with WPA3 security and segregates the attached staff/dept networks from the guest network. All designs will consider implementation's time issues with an NTP server which will ensure time keeping and devise proper centralized logging via a CISLOG server, in conjunction with compliance workflows will be controlled via strict conformity through a Web Application Firewall (WAF) to facilitate digital forensics, which is a must in attempting to catch and analyze security incidents. Finally, the developed proposal will be fully tested in a controlled GNS3 environment to reduce future problems and confirmation based on expedience. Once proven correct, deployment for the organization network will be in accordance with the implementation of the Global Security Compliance Program (GSCP), as well as the General Data Protection Regulations (GDPR) for operational security, as well as for individuals' privacy. Beyond the organizations policy for network security, the proposal considers the organization’s existing and maintained network, which will from there be standardized, scalable and compliant.

Keywords: Network redesign, VLAN segmentation, cybersecurity, firewalls, Wi-Fi security, GDPR compliance, digital forensics.

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Glossary of Terms

- VLAN (Virtual Local Area Network)
  - A network configuration that segments a physical network into smaller logical networks to improve security, reduce broadcast traffic, and simplify management.
- Firewall
  - A network security system designed to monitor and control incoming and outgoing network traffic based on predefined security rules to prevent unauthorized access.
- Intrusion Detection System (IDS)
  - A system that monitors network traffic for signs of suspicious activity and potential threats, alerting administrators when such activity is detected.
- Intrusion Prevention System (IPS)
  - A system that not only detects potential threats but also takes proactive measures to prevent or block malicious activities in real-time.
- GNS3 (Graphical Network Simulator-3)
  - A network simulation software that allows for the design and testing of network topologies, including routers, switches, and firewalls, without the need for physical hardware.
- Cisco Packet Tracer
  - A network simulation and visualization tool that helps in designing and testing network configurations, often used for educational purposes.
- Digital Forensics
  - The practice of collecting, preserving, analyzing, and presenting data from digital devices for the purpose of investigating security incidents or cybercrimes.
- Penetration Testing
  - A security testing method in which simulated attacks are performed on a system or network to identify vulnerabilities and weaknesses.

- GDPR (General Data Protection Regulation)
  - A regulation in the European Union that governs data protection and privacy, particularly for individuals within the EU. It sets strict guidelines on how personal data should be handled, stored, and shared.
- Access Control
  - A security feature that manages who can access network resources and what actions they can perform based on predefined rules or policies.
- Encryption
  - The process of encoding data so that it can only be accessed by authorized users, ensuring data privacy and protection from unauthorized access during transmission or storage.
- Scalability
  - The ability of a network or system to handle increasing amounts of work or traffic, and its capability to grow without compromising performance or reliability.
- Resilience
  - The ability of a network to maintain functionality and recover quickly from disruptions, such as cyber-attacks or system failures.
- IDS/IPS Integration
  - The combination of Intrusion Detection Systems and Intrusion Prevention Systems to both monitor and mitigate potential security threats in real-time.
- LAN (Local Area Network)
  - A network that connects computers and devices within a relatively small geographic area, such as an office building or campus, allowing for resource sharing and communication.
- Wi-Fi Segmentation
  - The process of dividing a wireless network into multiple segments (E.g.- guests and staff Wi-Fi), ensuring that sensitive resources are not accessible from guest networks.
- Ransomware

- A type of malicious software that encrypts a victim's data and demands payment (ransom) for the decryption key.

❖ Phishing

- A cyber-attack in which attackers deceive individuals into providing sensitive information (E.g.- passwords, credit card numbers) by pretending to be a trustworthy entity.

❖ Data Breach

- An incident where unauthorized individuals gain access to sensitive or confidential data, often resulting in data theft or loss.

1. Introduction & Literature Review

1.1. Introduction

Every year there is increased urgency behind the need to secure network infrastructures as the landscape of threats towards cybersecurity has changed - and grown in complexity and scale. Network security is no longer simply protecting data but also protecting the reputation of the company, protecting its intellectual property, ensuring business continuity, etc. The project here is to redesign the network infrastructure for ABC Holdings PLC - a company with a complex organizational structure where they operate across many different departments. The objective of the project is going to be based around an existing network which can't handle the new challenges involved with the requirements of cybersecurity. In the fit-out of the building, there are eight departments, and a majority of the infrastructure is located on the ground floor, which means a fresh approach to every aspect of network infrastructure used by the business. Especially given that there are no communications controls between the VLANs, either for LAN connections or Wi-Fi connections. The main server is also exposed to physical security risks. To address these challenges, this project aims to redesign the network, incorporating VLAN segmentation, firewalls, IDS/IPS systems, and secure Wi-Fi. Additionally, the design will include DNS, DHCP, FTP, and Syslog servers to improve network management and security.

1.2. Background and Motivation

ABC Holdings PLC is now operating with several security vulnerabilities due to the outdated network design. With the network containing only two VLANs and no security features, unauthorized users can gain access to the network and create major complications like data breaches. There are also internal security breaches as the server is located on the ground floor, there is a small likelihood of physical breaches on the server. The departments are not separated well enough in the network leaving them liable to exposure of sensitive data to unauthorized users, or even a worst-case scenario, malicious actors.

The driving force for this project is new network technology, an increasing concern with the company's network regarding data confidentiality, integrity and availability from potential loss or destruction. The growing number of cyber security threats such as ransomware, phishing attacks and the potential loss of data through ex filtration will all greatly affect the integrity and confidentiality of the organization from protecting sensitive or regulated data. Therefore, the driving force of this project is to design, implement, and manage a fully secure network infrastructure that mitigates these risks through the use of new security technology and security best practices. The objective of the project is to increase the growth of the network and the potential to grow, while also becoming more resistant to cyber-attacks without understanding. In

addition to enhancing the organization's ability to monitor, detect and respond to a cyber security incident through the use of digital forensics technology.

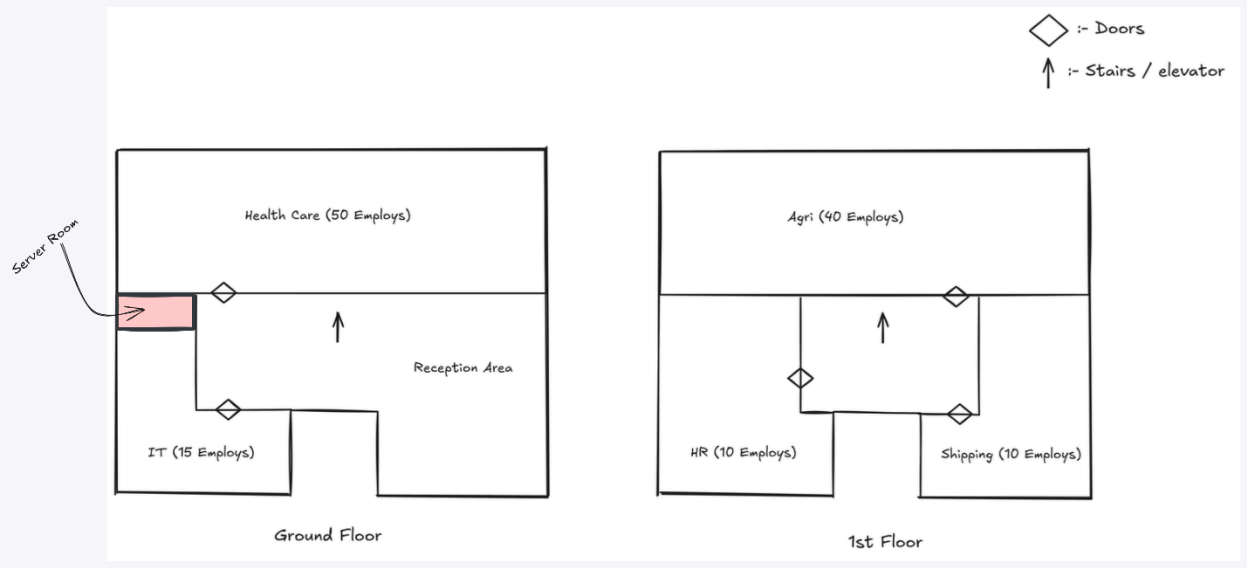
Previous research has identified best practices in network securities, for example, utilizing layered security architectures to combat internal and external attacks. Layered security architecture technology includes firewalls, Intrusion Detection Systems (IDS) and Intrusion Prevention Systems (IPS). Digital forensics contributes to monitoring and managing incident response, supplying organizations with evidence that can be collected, preserved, and subsequently examined after an incident to establish preventative actions in the future. This project will leverage virtualization technology, GNS3 and Cisco Packet Tracer, to develop the network design and perform robust testing across a range of potential cyber threats.

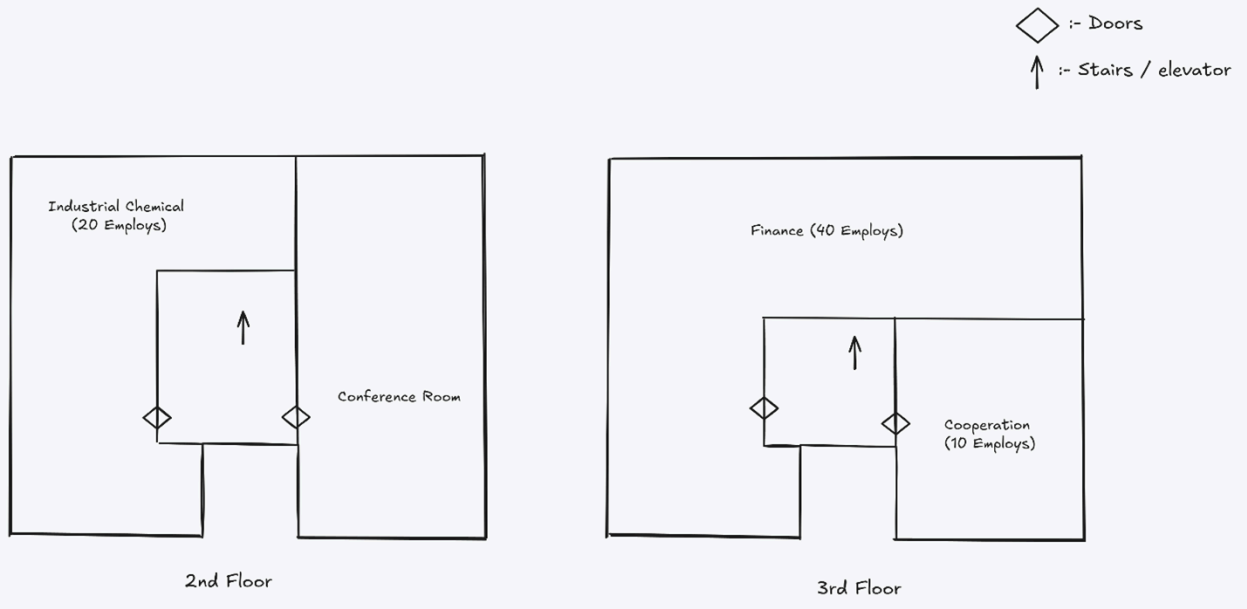
The main elements of security technology, firewall, IDS/IPS and digital forensics will all be instruments of consideration and be included as part of the network design. While the project has a focus on security technology, it will not only result in a secure network infrastructure for ABC Holdings PLC, but it will also provide an exemplary for future network design that considers security and resilience.

1.3. Problem in brief

ABC Holdings PLC currently employs a network design that no longer meets the security and organizational needs and is associated with its multi-departmental needs. The company has designed its network with only two VLANs, one for LAN connections, and one for Wi-Fi. The LAN VLAN connects all of the department(s) to a central hub server and does not have any safeguards that control communication from the central hub or directly to sensitive resources or services between departments. In other words, all departments can communicate directly from the LAN to each other and allow access to an internal threat with no limitations to the critical systems and data of the company. In addition, there are no network-based safeguards preventing departments from accessing other department information, creating an opportunity for an exploited environment for any malicious attacker. The Wi-Fi VLAN has two segments, Guest Wi-Fi and Staff Wi-Fi; however, there are no communication safeguards established between the two. The staff and guest networks are separate, but users on the guest network might have the ability to access staff-networked sensitive information without any safeguards to appropriately encrypt the data, or without specifically authored firewall rules. Only the social media sites are restricted when using either of two segmented Wi-Fi networks. This security measure does not mitigate the broader risks of unauthorized access to sensitive data.

It does seem that the network lacks any notable security features such as a lack of VLAN separation, firewalls, and IDS systems. With four levels and eight divisions in the building, this is rather alarming. There are no means of secure communication, no network segregation, and any sensitive information is easily accessible to employees or visitors who do not have that access. In addition, the central server stores sensitive organizational information and is also located on the first floor with the IT office which exposes the server to potential physical security issues, especially in a building with poor access controls. The network is no longer conducive to quality security as it creates many vulnerabilities to the Department of Human Services (DHS) consistent of internal breaches of information, exposing sensitive information to unauthorized individuals, and greater exposures to cyber-attacks from internal and external individuals. A redesign would be needed to include segmentation, secure communications across departments, and a variety of security measures to guard against identified risks. Without continued investment in the network, ABC Holdings PLC is still vulnerable to the pervasive issues mentioned above.


:- Doors
:- Stairs / elevator
Server Room
Health Care (50 Employs)
Reception Area
IT (15 Employs)
Ground Floor
Agri (40 Employs)
HR (10 Employs)
Shipping (10 Employs)
1st Floor


:- Doors
:- Stairs / elevator
Industrial Chemical
(20 Employs)
Conference Room
2nd Floor
Finance (40 Employs)
Cooperation
(10 Employs)
3rd Floor

1.4. Aim & Objectives

1.1.1. *Aim*

Developed and implemented a secure network architecture for CIG Holdings PLC in GNS3. Incorporated firewalls, IDS/IPS, and cyber forensics to mitigate the risk of internal and external threats to data, as well as provide business continuity in the event of a cyber threat.

1.1.2. *Objectives*

- To conduct a comprehensive review of the existing network security challenges and vulnerabilities at CIG Holdings PLC.
- To study and evaluate technologies, such as firewalls, IDS/IPS, VLAN segmentation, and digital forensics, to identify effective solutions for improving network security.
- To design and develop secure network architecture with proper VLAN segmentation, secure communication protocols, and robust security measures.
- To simulate the network design using tools like GNS3 and Cisco Packet Tracer to test and evaluate the security features and performance of the new infrastructure.
- To document the design process, implementation details, and testing results, providing a comprehensive report for future network improvement and security evaluation.

1.5. Scope

1.1.3. *Inclusions*

- **Network Redesign**
  - The project will redesign ABC Holdings PLC's existing network infrastructure to improve its security and performance. This includes implementing VLAN segmentation, secure communication protocols, benefiting from firewall, IDS/IPS, and other cybersecurity capabilities.
- **Simulation and Testing**
  - The network design will be simulated using tools like GNS3 and Cisco Packet Tracer to test the effectiveness of the design and identify any vulnerabilities or weaknesses.
- **Security Features**
  - The scope includes implementations of security features such as encryption, access controls, and secure wireless networks (WPA3), to lessen the risk of weak passwords and to maintain sensitive data from unauthorized access.
- **Compliance**
  - The project will develop the network for the design to comply with the appropriate security legislation such as GDPR and Global Security Compliance Program (GSCP).
- **Digital Forensics**
  - The design would apply digital forensic capability through centralized logging, intrusion detection, preserving evidence, etc. which may help with the detection/analysis of security incidents if ever needed.

1.1.4. *Exclusions*

- **Hardware Implementation**
  - The project will not involve the physical installation or configuration of the network infrastructure at ABC Holdings PLC. It is focused on designing and simulating the network for testing purposes.
- **Third-party Vendor Integration**
  - It is outside the scope of this project to integrate with external vendors or services outside of the security technologies noted (e.g., vendor-specific hardware).
- **End-User Training**
  - The project documents will provide information, but there is no intention to provide training on best practices for end-user training on network security best practices as part of this project.
- **Post-Implementation Monitoring**

- Ongoing monitoring and maintenance after the deployment of the network redesign will not be included in this project.

1.6. Deliverables

1. **Network Design Documentation**

The network design documentation will specify the new network infrastructure for ABC Holdings PLC. This includes identifying VLANs, firewall configurations and IDS/IPS controls to be implemented. The network design documentation is descriptive and covers the architecture of the network design with security provisions in place by outlining a 'secure and efficient' operation across the departments of the organization.

2. **Simulation Results**

The simulation report, which will test the operational design using simulations for the design using tools like GNS3 and Cisco Packet Tracer, will be included as part of this delivery. The report will outline the network performance and simulation of the security provisions which should identify vulnerabilities and recommendations for improvements prior to implementation.

3. **Implementation Plan**

The implementation plan will provide a detailed breakdown on how to activate the new and redesigned network infrastructure which will include configuration of VLANs and firewall settings, and the installation of IDS/IPS system. The implementation plan should include a deployment timeline, allocation of resources assigned, and a risk management plan should include an implementation approach as well as considerations to apply for compliance with security regulation requirements.

4. **Final Project Report**

The final project report will describe the project from start to finish, including the successful delivery of the objectives and overcoming challenges to complete the project. To conclude this report there will be recommendations to manage the networks' future security and performance.

1.7. Literature Review

1.1.5. Introduction

The era of digital transformation has shifted the landscape in organizations across the world. As organizations depend on information technology (IT) systems to carry out day-to-day business, network security emerged as one of the most critical issues for organizational asset protection. As organizations become more digitized and interconnected, the threat of cyber security continues to grow, which emphasizes the importance of having robust network security frameworks to protect the confidentiality, integrity, and availability of sensitive information. The cost of cybercrime globally is estimated to reach USD 10.5 trillion per year by 2025 (*Canadian Centre for Cyber Security, 2023*), which provides a glimpse of the depth of the threats organizations face worldwide. For organizations like ABC Holdings PLC, a diversified organization with multiple divisions each with access to various sensitive data assets, security planning will certainly be a pressing issue. Organizations must adequately prepare for data breaches, ransomware, and unauthorized access as they may result financial losses, damaged reputation which may disrupt customer relations, and legal liabilities. In its 2021 report, IBM stated the average Global cost of a data breach is about USD 4.24 million (*IBM, 2021*). This highlights how important it is to develop a proactive cybersecurity plan.

The objective of this literature review is to identify core strategies and technologies required for the purpose of designing a secured network for ABC Holdings PLC, by employing VLAN segmentation, firewall configurations, IDS/IPS systems, and digital forensics with the end goal of ensuring the company's network infrastructure can mitigate external and internal cyber threats, while the company monitors, detects, and responds to security events and incidents. As cyber threats evolve, understanding current network security technologies and following best practices are vital for ensuring the sustainable future of ABC Holdings PLC's network resilience.

1.1.6. Cybersecurity Awareness and Threat Landscape

Awareness of cybersecurity is an important first step to reducing the risk of cyber threats because human error is one of the biggest risks that any organization has to manage. The more aware employees are and the more training they receive about possible cyber threats, the less likely an organization is to fall victim to phishing, social engineering, and malware. Alvitigala and Mithrananda (*Alvitigala and Mithrananda, 2024, p. 4*) report that SMEs in Sri Lanka, much like many parts of the world, are highly concerned about cyber threats but severely challenged in implementing meaningful cybersecurity measures. The main reasons are limited resources and limited awareness of dangers imposed by cybercriminals. While larger organizations tend to have more available resources for training on cyber hygiene, such as identifying suspicious emails or how to manage passwords, employees may not receive adequate training on these topics,

creating risk for their employer. The importance of employee awareness is critically important for organizations dealing with multiple departments such as ABC Holdings PLC, where numerous departments manage sensitive information. Providing regular employee training on topics such as phishing awareness, password management, along with an organizational cybersecurity culture will greatly address reducing the possibilities of human error causing breaches. While it is important to teach employees the technical aspects of cyber security, it is also important to help employees understand the broader threat landscape.

The cyber threat environment has dramatically transformed over the last decade. Attack vectors have become increasingly more complicated, and attacks including ransomware, DDoS attacks, and data breaches, have been pressing organizations worldwide. For example, the Colonial Pipeline ransomware attack in 2021 impacted fuel distribution across parts of the U.S., demonstrating the widespread impacts of a cyberattack on critical infrastructure (*Lavrov, 2021*). These attacks can highlight a dual approach within organizations which reinforced technological defenses but helped develop a security culture that embraces security across every level of the organization.

Government is important for improvement of cyber security practice. In Sri Lanka the government have frameworks and steps such as Sri Lanka CERT|CC and the Cybersecurity Regulatory Authority to promote national Cybersecurity awareness (*Livera, 2025, p. 6*). For ABC Holdings PLC, developing policies internally paired with national regulations, can cement the external defenses against cyber threats, increase the organizations' overall cybersecurity resilience, compliance with cybersecurity laws, regulations for the industry, and GDPR or PCI-DSS.

1.1.7. VLAN Segmentation and Network Security

VLANs, or Virtual Local Area Networks, are important components of securing and optimizing current networking frameworks. By clustering segments of a larger network into smaller groups, VLANs can provide efficiencies of performance and improve security by continuing to segregate different types of traffic. A VLAN groups similar departments, devices, or applications when performing administrative duties, e.g., critical User/Payroll data may be segregated from other users on a network. ABC Holdings PLC can ensure their payroll functions of finance and HR are as separate or as heavily imposed permissioned and controlled as necessary; thus, there is better security if there are fewer users on a VLAN. There are multiple dimensions of improved controls to separate departments, applications, devices, etc. The critical command is "less is more" in order to reduce the surface area for attacks on exposed networked assets and the likelihood of possible attacks. A breach of one part of the network will have reduced ramifications on other VLANs and risk from unauthorized actions. In his case study, Perri (Perri, 2003, p. 7) participates in discussions around internal attack potentials and designs events on VLANs that allowed an administrator to assure that a critical computer system would

be isolated from a less secure registered trader insistence. VLANs provide assurance boundaries for traffic that passes between departments or devices, and if ABC Holdings PLC segmented their network into security zones for its guest networks for external networks and its internal networks focused on employees, they could limit unauthorized access to critical systems as well as inadvertently leaking data in predictable situations.

There are challenges associated with VLAN segmentation. VLAN misconfigurations, such as VLAN hopping or inappropriate routing, can limit the measures to block potential attacks for exploitation. Shaik (*Shaik, 2024*) recognizes that VLAN segmentation security measures come with tremendous benefits; however, they are only effective if configured and monitored appropriately. Network security practitioners at ABC Holdings PLC must ensure that the network infrastructure is designed and configured such that it securely segments and controls data flow - similar to the two examples given above in which ACLs and firewalls, may enforce access concepts. Furthermore, new technologies are emerging such as Software Defined Networks (SDN) or VXLAN (Virtual Extensible LAN) that build upon VLANs by providing increasingly scalable and flexible solutions. Both technologies support rapid network growth, have simplified traffic management, and supported the development of secure networks in organizations with long term plans for future implementations, which make them applicable to ABC Holdings PLC (Shaik, 2024, p. 10) .

1.1.8. Firewall Technologies and IDS/IPS Integration

Firewalls are the primary defense against external threat actors and prevent unauthorized access to network resources. A firewall acts like a very advanced traffic guard, allowing and denying incoming and outgoing network traffic in real-time, based upon defined security policies. Firewalls come in both software and hardware versions. Similarly, firewalls can either be stateless or stateful. A stateful firewall observes the state of active traffic flows, which enables a firewall to be meaningful about the connection allowing access when the traffic and prior connection references allow it (*Wang, 2022, p. 5*). Firewalls can include many advanced features. Wang (*Wang, 2022, p. 5*), describes next-generation firewalls (NGFW) which include advanced capabilities beyond traditional firewall functions, such as, deep packet inspection, application layer filtering, and intrusion prevention to protect against the advanced persistent threat (APT). Of course, firewalls can also integrate with Intrusion Detection Systems (IDS) and Intrusion Prevention Systems (IPS) to create an even more robust defense in depth framework on the network. The additional functionality of IDS and IPS systems can provide response capabilities for incoming threat detection. An IDS monitors and uses the traffic by asking a series of questions to identify suspicious activity on the network, while an IPS can automatically block any malicious traffic based on pre-defined security policies. Chiba et al. (*Chiba et al.,2022, p. 13*) states that when firewalls integrate an IDS and an IPS, it becomes an integrated system that is designed to counter network threats both from internal and external threat actors, using a layered defense structure with dedicated, real-time, layered threat protection. By combining these

systems, ABC Holdings PLC can now implement an end-to-end solution that detects, prevents, and when necessary, reacts to potential malicious activity so that a potential issue is identified and addressed in an effective manner before major damage is caused. The maintenance of firewalls, as well as the configuration, maintenance, and monitoring of an IDS/IPS system rely heavily on timely monitoring, examining potential issues, and analysing security logs. As Perri mentions (*Perri 2003, p. 8*), traditional firewalls and IDS/IPS solutions were designed to address security at a single given time and one of their weaknesses is the alert fatigue that comes from alerting these many levels resulting in huge amounts of logs to sort through. For ABC Holdings PLC, setting up a centralized logging solution and employing a Security Information and Event Management (SIEM) system to analyze potential issues will help mitigate this alert fatigue and provide an oversight function for security incidents where action must be taken in a timely manner.

1.1.9. Digital Forensics and Incident Response

Digital forensics has important applications in contemporary network security, particularly in the context of incident response and cyber investigation. Simply stated, digital forensics involves the collection, preservation, and analysis of digital evidence located on networks, devices and systems so it can be used in court. Digital forensics is very powerful way to determine who carried out an attack and how an attack was executed. Digital forensics can also help organizations understand the total extent of the damage to their organization and preserve evidence for future litigation (*Qureshi et al., 2021, p. 9*). On the other hand, incident response (IR) relates to acting upon a breach and dealing with the subsequent activities. The overall goal of an IR plan is to limit damage from an incident and identify and contain the data breach, then return systems to normal operation in the shortest time possible. Liu et al. (*Liu, Singhal and Wijesekera, n.d., p. 11*) propose the importance of viable incident response plan is to have a clear definition of detecting, containing, eradicating, and recovering from IR incidents. ABC Holdings PLC need to ensure forensic readiness is noted in the network design so they can detect an intrusion and start collecting evidence immediately without compromising the data.

The incident response lifecycle consists of several phases, each of which plays a critical role in mitigating the effects of a breach,

- Detection: Identifying the occurrence of a security incident.
- Containment: Limiting the scope of the damage to prevent further spread.
- Eradication: Removing the threat and ensuring vulnerability is addressed.
- Recovery: Restoring affected systems and resuming business operations.
- Lessons Learned: Analyzing the incident to prevent similar breaches in the future.

For ABC Holdings PLC, it is essential that they create an incident response team (IRT) as well as invest in forensic tools that can be integrated with their existing network infrastructure.

There are various forensic tools such as EnCase, FTK, and Autopsy, which ultimately offer companies the ability to collect and gather data from compromised machines and sources and analyze that data to seek out malicious activities. The importance of chain of custody in terms of evidence is paramount in terms of admissibility in court (*Perri 2003, p. 8*), especially if ABC Holdings PLC pursues any legal action against the perpetrators in the future. The Liu et al., (*Liu, Singhal and Wijesekera, n.d p. 13*), paper provided a proposed logic-based approach in the way to analyze the evidential forensic material as it creates a framework based on inductive reasoning-based stock accounts and vulnerability databases to reveal the pathways followed to reconstruct any attack scenario. This allows security teams to understand what methods the attackers may have used, what vulnerabilities they exploited and to obtain a much clearer picture of what transpired and develop remediation action related to the breach. The article also emphasized the need to reduce forensic analysis, so that organizations can respond to a threat faster in high-stress environments. The problem of encrypted data and anti-forensic tools is a substantial barrier in incident response. In today's cybercriminal landscape, attackers regularly deploy techniques to erase indications of an incident such as data-wiping, as well as log modification/dismissal. A preventative solution would be for ABC Holdings PLC to implement a firm, established forensic policy, to favored, to make and procure disks/images, preserve tamper evident data, and use forensic tools and applications that can bypass common anti-forensic techniques. Also, with advancing forensics technology and AI driven tools, tools will continue to evolve to limit the severity and time needed to assess significant datasets while continually understanding the underlying cause(s) for the attack.

1.1.10. Emerging Technologies in Network Security

As the cybersecurity landscape continues to evolve, emerging technologies are playing an increasingly vital role in enhancing network security. Threats from modern cybercriminals are sophisticated and dynamic; therefore, organizations must rely on more advanced ways to secure their network and data from potential breaches. Organizations are executing a plan by adopting emerging technologies, such as Artificial Intelligence (AI), Machine Learning (ML), and Blockchain that enhance their security infrastructures and provide capabilities to identify, mitigate, and respond to a cyber threat as real-time security happens. Artificial Intelligence and Machine Learning are changing how organizations detect cyber threats. AI can use algorithms to recognize patterns and predict vulnerabilities in devices connected to the organization's network. Many legacy systems retain their security risks because they create signatures or rules that identify network vulnerabilities or the attack on a known rule. By leveraging AI/ML capabilities, organizations can capitalize on their potential to address unknown and dynamic threats. According to Admass, Munaye, and Diro (*Diro, 2024, p. 5*), AI based intrusion detection system (IDS) autonomously identifies irregular behaviors with network traffic, such as access at transactional time of the day does not associate with user access or unauthorized access patterns.

The IDS learns and improves a organizations' ability to recognize past threats while simultaneously blocking an emerging attack pattern.

In the case of ABC Holdings PLC, AI-powered IDS/IPS systems would vastly improve the organization's ability to identify threats using near real-time analyses and automated response apparatuses to address threats at a velocity faster than conventional mitigation plans can provide, including time-frames that eliminate the timeline for addressing zero-day attacks and insider threats by authoritatively eliminating them as risks prior to their expansion into a full-throttle attack. Blockchain as a technology is also particularly relevant to mitigating cyber security risks while supporting the protection of data integrity and transaction verification. The distributive power of the blockchain perimeter instead of traditional perimeter-based frameworks technology, combined with the immutable technology for solidifying data authenticity puts blockchain in the forefront for augmenting data security associated with secure transactions which is paramount in many vectors of cyber aspects. For example, leveraging blockchain in the supply chain, ensuring authenticity of transactions, and avoiding fraudulent transactions of data in conventional UA(N) and UA(D)-level network systems are just a few possibilities. Although the cybersecurity industry has not yet seen widespread use of blockchain for even general network security, the potential of blockchain technology to enhance the future of network security related to identity management, access control, and data validation may change many network security standards in the future (Shaik, 2024, p. 14). For ABC Holdings PLC, blockchain can be explored for use in facilitating secure transactions, but mobile access control could be achieved by using blockchain to facilitate secure access to data, contracting and reporting to validate integrity over time. Smart contracts combined with blockchain-based access control could also be implemented to improve organizational data protection policies as warranted.

Another important developing technology is Zero Trust Architecture (ZTA). The Zero Trust model presumes that everyone attempting access to the network (both internal and external to the organization) is a potential threat until proven harmless. In the age of remote work and cloud computing, ZTA is more important than ever to facilitate that the user, device, and application accessing the network has authenticated their identity and has confirmed their security posture actually allows them access (*Livera, 2025, p. 8*). For ABC Holdings PLC, Zero Trust would address the challenge of remote workers, contractors and external vendors accessing sensitive data. As part of the Zero Trust model, utilizing technology such as multi-factor authentication (MFA), micro-segmentation and continuous monitoring of access means that no user or device can be trusted by default. Utilization of Zero Trust technology throughout the ABC Holdings PLC network would drastically reduce revenues for unauthorized access, and that would also enhance the resiliency of the network itself.

1.1.11. Future Directions and Challenges in Network Security

Network security technologies will continue to develop, however, new problems will develop, especially if their environments are complex and large scale like that of ABC Holdings PLC. One of the challenges which Admass, Munaye, Diro (*Admass, Munaye and Diro, 2024, p. 7*) raised was the increased sophistication of cyberattacks which now must be addressed using a defense in depth approach which requires organizations to consider technical cyber defense and the human factor. A further challenge being experienced by many institutions is a global shortage in cybersecurity professionals. As cyber threats get more sophisticated there is a growing demand for highly skilled cyber professionals to develop and employ cyber defenses. The (ISC)² Cybersecurity Workforce Study found that globally, the cyber security profession faces a worker shortfall of 3.5 million. (ISC)², (2021). For ABC Holdings PLC this will mean supplementing their internal teams and employing managed security service providers (MSSPs) to fill an expertise gap.

Another problem is putting together various security technologies into an integrated, cohesive, and efficient model. For example, when combining the technologies of AI/ML systems, firewall systems, IDS/IPS Systems, and network segmentation, the compliance of each technology needs to be reviewed and planned so it runs correctly and does not add to the incidence response burden or alarm fatigue, that could overwhelm an administrator with duplicative security controls or alerts. Shaik (*Shaik, 2024, p. 13*) describes that automating streamlining practices in network security can aid cybersecurity metric reporting and action plans, will help alleviate human tolerance errors, and will ultimately allow a faster response to a security incident. Beyond automation and actioning compromised security measures/compliant activity, organizations such as ABC Holdings PLC will want to start preparing for the 5G and Internet of Things (IoT) that have become more prevalent and appear to be rising exponentially. The vast number of devices connected to the internet worldwide creates a layer of complexity and attack surface to breach the commonly held belief of high-speed data coupled with getting data from point A to point B while demanding accountability of that data. Increasingly we will be required to secure infrastructure for 5G IoT devices that will become conscientious to validate security of IoT devices routinely. The implementation of SDN and VXLAN will be essential to ensure we can enact and facilitate technological changes with scalable and secure networking options (*Shaik, 2024, p. 15*). Of course, to stay current and become a valued pioneer and strategic leader today and ahead of current will not only benefit the future of SALC and ABC Holdings PLC's business strategy and viability but more importantly one of the greatest accomplishments during initial phases of connection security than was last done 5 years ago in 2018. Organizations need to routinely review their security posture and organizational resources, provide training to employees, and invest in proactive means of defense so they possess the ability to detect, defend, and respond to a threat prior to an incident or breach.

1.1.12. Conclusion

In conclusion, to secure a corporate network, an approach that encompasses multiple layers of security, leveraging technology, best practices, and proactive measures is required. For ABC Holdings PLC, building a secure network infrastructure with the inclusion of VLAN segmentation, firewalls for internal and external traffic, IDS/IPS systems, Zero Trust (ZT) models, and digital forensics capabilities will maximize its capability to prevent, detect and respond to cyber threats. In addition, it will be necessary to leverage emerging technologies such as AI/ML, blockchain technology and secure 5G in support of the organization's security posture in the future. Given the pace of change in cyber threats and risks to organizations, it is critical for organizations to have the capacity to enable their cybersecurity framework to be flexible, adaptable and resilient against a range of threats and risks. In summary, the incorporation of advanced technological capabilities and procedures, coupled with a strong cybersecurity culture and clearly defined incident response procedures, will enable ABC Holdings PLC to ensure its critical assets are safe, adhere to industry regulations, and maintain the trust of its clients and other stakeholders.

2. Analysis

2.1. Problem Definition

ABC Holdings PLC's present network architecture has a number of critical vulnerabilities stemming from a lack of VLAN segmentation, lack of Wi-Fi security, and a lack of internal communication controls. The architecture has two VLANs; one for LAN connections, and one for Wi-Fi. This currently does not segregate sensitive data or restrict access between different departments, ultimately exposing the organization to internal and external threats, unauthorized access, and advanced data exfiltration. In addition, the main server, which contains critical organizational data, is located in an area with limited physical security, increasing the risk of a potential breach. Also, there are no backup systems in place for the ISP, firewall, or main switches, which would leave the network vulnerable to downtime if the hardware fails or were to suffer issues with external connectivity.

2.2. Analysis Techniques

To understand the weaknesses of the current network infrastructure, several analysis techniques were employed,

- SWOT Analysis
    - This helped identify the strengths, weaknesses, opportunities, and threats within the current network design. It provided a comprehensive view of where improvements were needed.
- Use Case Analysis
    - This involved defining specific use cases for each department and user group (e.g., Finance, HR, guest Wi-Fi) to determine their access requirements and how best to control traffic between them.
- Requirement Engineering
    - A thorough gathering of business and technical requirements was conducted to understand the network's current limitations and the company's future needs in terms of scalability, security, and performance.

2.3. Key Findings

The analysis revealed several issues in the current network design.

- Lack of VLAN Segmentation.
    - The absence of proper VLAN segmentation allows departments to communicate freely with each other, putting sensitive data at risk.
- Insufficient Wi-Fi Security
    - The guest and staff Wi-Fi networks are poorly segregated, with no safeguards preventing unauthorized access to sensitive data.
- Weak Physical Security
    - The server's location in an area with limited physical security increases the risk of unauthorized access to critical systems and data.
- Lack of Incident Response Mechanisms
    - There are no mechanisms in place for detecting, responding to, or mitigating security incidents, leaving the network vulnerable to attacks like ransomware or phishing.
- No Backup Systems
    - The network currently lacks backup systems for key infrastructure components, including the ISP, firewall, and main switches, which could lead to significant downtime if any of these components fail.

2.4. Conclusions from Analysis

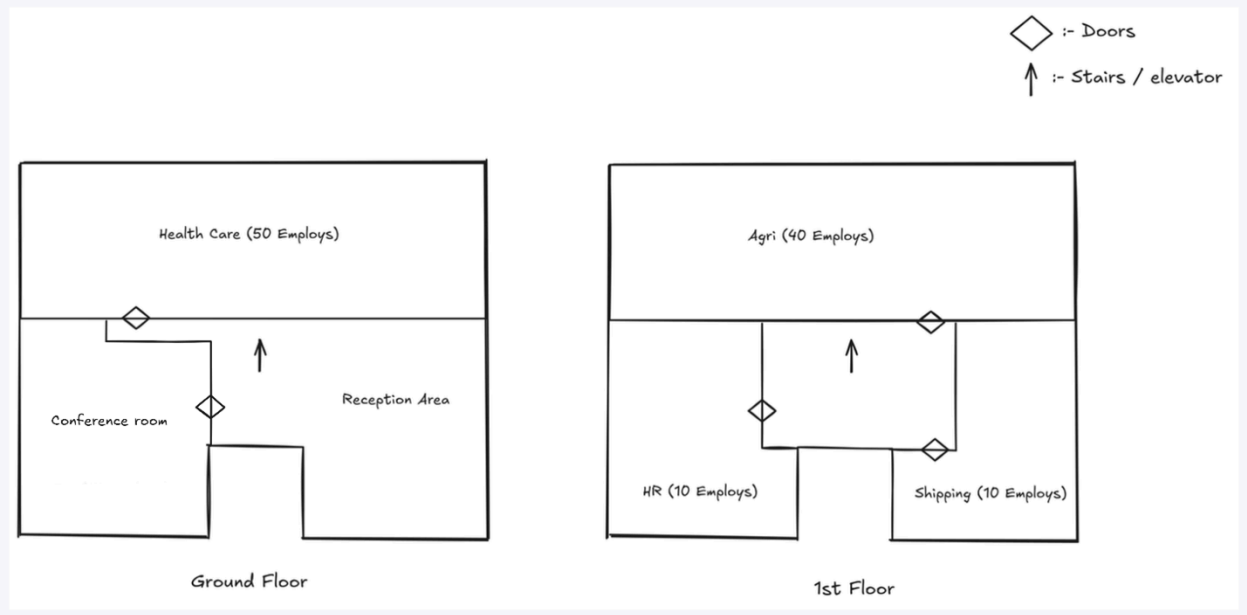
The existing network system at ABC Holdings PLC is outdated and incapable of supporting the organization's security and operational needs. There is no segmentation of network devices, poor wireless security, and physical vulnerabilities which threaten sensitive data and business functions. Moreover, there is no backup for critical infrastructure systems. A new networking design, including VLAN segmentation of devices, firewall placement, IDS/IPS, security improvements, and backup systems, is required to provide a secure network and a guarantee of continuity of services within the organization, even when hardware failures or other threats occur.

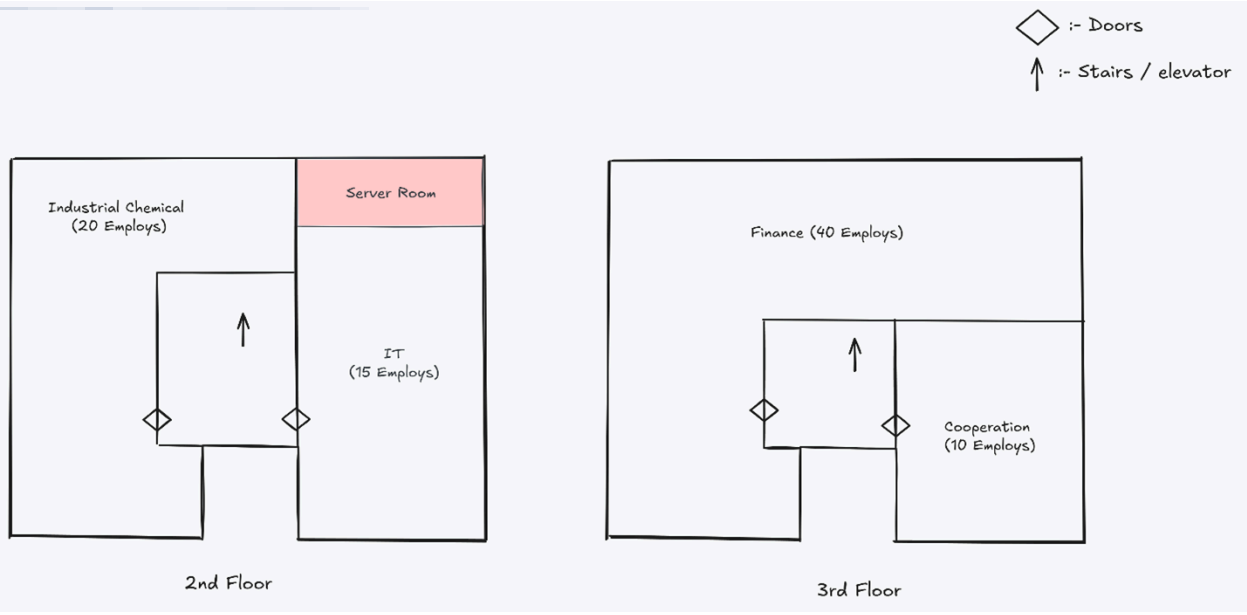
3. Design

3.1. Design Techniques

The ABC Holdings PLC network design incorporates advanced techniques to achieve security, scalability, and operational efficiency.

- VLAN Segmentation
  - Each department utilizes its own VLAN to provide secure isolation of sensitive data and minimize the attack surface and data flows between system departments. Inter-VLAN communication is controlled by firewalls.
- Firewall Configuration
  - FortiGate firewalls are deployed at critical points in the network to control traffic and provide protection against the internal and external threat landscape. The firewalls to help prevent security breaches are also leveraged with advanced features like deep packet inspection and threat intelligence.
- Load Balancing
  - A load balancing solution is deployed to improve the availability and reliability of all critical applications. This helps accomplish even distribution of the network traffic across all servers, avoids overload, and improves performance.
- Access Control
  - Role-based access control (RBAC) is implemented to limit access to sensitive resources based on user roles. This assures only authorized persons have access to critical data and systems and limits the risks of insider threats.
- Redundant Systems
  - To maintain a high level of availability and redundancy, the network provides backup systems for the ISP, firewall, and core switches. If one goes down, it automatically switches to the backup, enabling continuity of business operations and less unproductive downtime.
- IDS/IPS
  - The network has intrusion detection and prevention systems (IDS/IPS) that monitor the traffic for suspicious activity. The IDS/IPS systems block the malicious traffic automatically. The IDS/IPS provides additional levels of security to the overall system.


:- Doors
:- Stairs / elevator
Health Care (50 Employs)
Conference room
Reception Area
Ground Floor
Agri (40 Employs)
HR (10 Employs)
Shipping (10 Employs)
1st Floor


:- Doors
:- Stairs / elevator
Industrial Chemical
(20 Employs)
Server Room
IT
(15 Employs)
2nd Floor
Finance (40 Employs)
Cooperation
(10 Employs)
3rd Floor

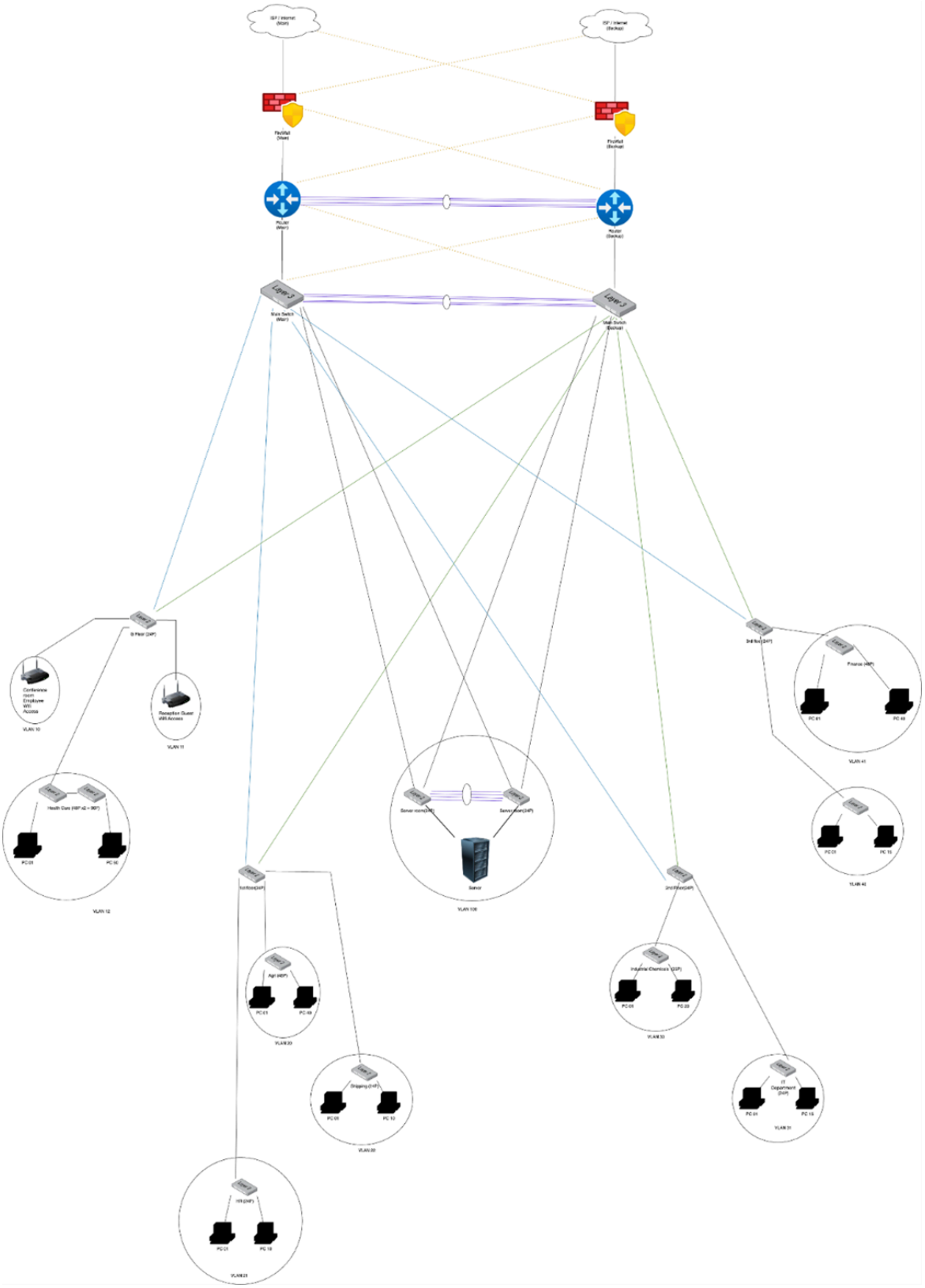


Figure 5High Level Network Architecture

3.2. System Overview

The structured layout of the network for ABC Holdings PLC is based on a multi layered security architecture, ensuring that data is secured at every tier.

- Core Network
    - The core network includes routers and switches that direct traffic across the different VLANs. Firewalls and IDS/IPS devices are used to secure the outside of the network and scan for threats.
- Protection of Servers and Data
    - There are critical servers, DNS, DHCP, FTP and Syslog residing in secure VLANs. These servers are isolated from user traffic and are defined to strict access controls to protect against the user accessing the isolated servers. The servers are further supported with backup components to ensure that operations continue.
- Workstations and Networked Devices
    - Workstation will be connected to the proper department VLANs; additional access controls are port security, access control lists, and monitoring no devices on the network. This component needs to be secure to limit an unapproved device from connecting to the wrong VLAN.
- Backup Systems
    - Backup components should be in place for ISP, firewall and core switches. This fixed physical environment will provide seamless connectivity when the existing physical component fails.

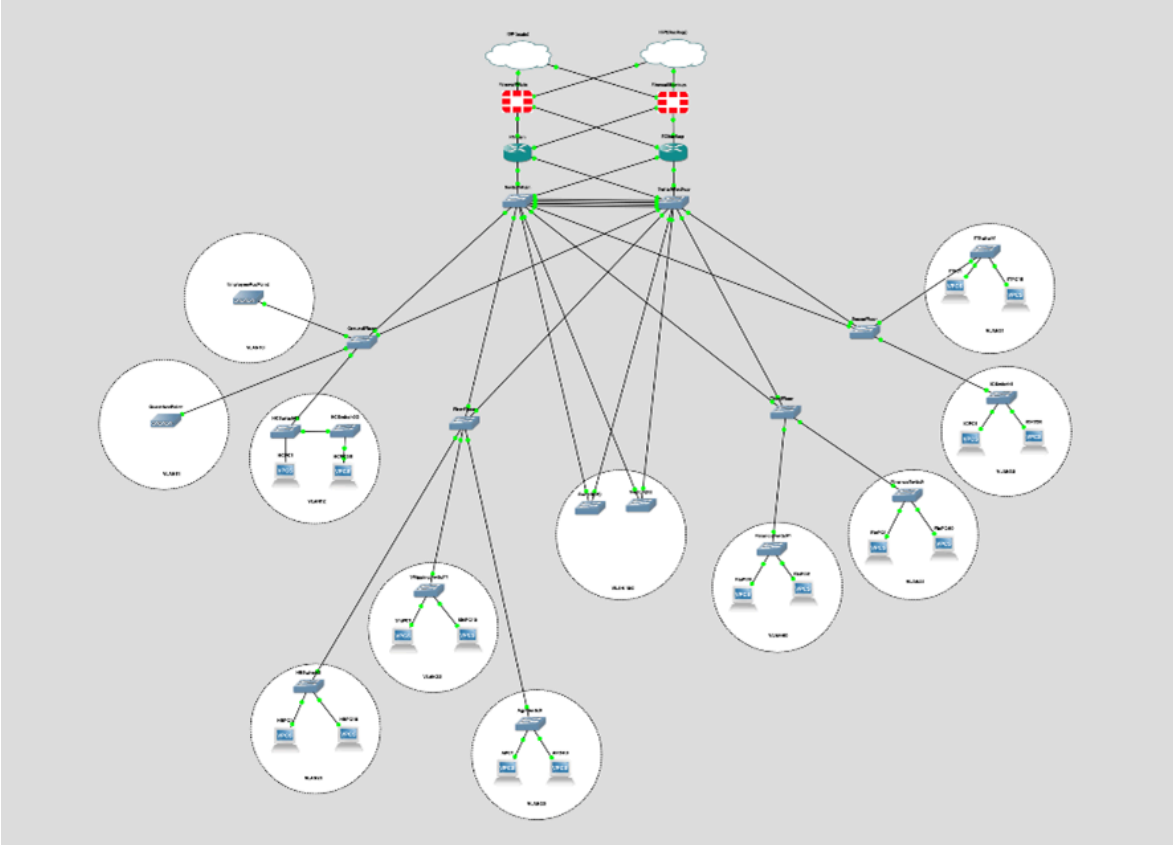


Figure 6network diagram Image

3.3. IP Configuration

The network is segmented into different VLANs, with each department allocated a specific range of IP addresses to ensure proper communication and isolation. Below is the IP configuration for the network.

Table 1 IP Conf

Host Bits	10
Subnet Mask	255.255.252.0
Network IP	192.168.0.0
Broadcast IP	192.168.2.255
Usable IP Count	1,022
Usable IP Range	192.168.0.1 - 192.168.2.254

Floor	Location/Department	Workstation count	VLAN	Assigned IP Range
Ground Floor	Health Care	100	VLAN 13	192.168.0.130 - 192.168.0.227
First Floor	Agri	80	VLAN 21	192.168.0.230 - 192.168.0.307
First Floor	HR	20	VLAN 22	192.168.0.310 - 192.168.0.327
First Floor	Shipping	20	VLAN 23	192.168.0.330 - 192.168.0.347
Second Floor	Industrial Chemicals	40	VLAN 31	192.168.0.350 - 192.168.0.387
Second Floor	IT Department	30	VLAN 32	192.168.0.390 - 192.168.0.417
Third Floor	Finance	90	VLAN 42	192.168.0.420 - 192.168.0.507
Third Floor	Cooperation	16	VLAN 41	192.168.0.510 - 192.168.0.523
		AccessPoint Count		
Ground Floor	Conference Room (Employee WiFi)	2	VLAN 11	192.168.0.81 - 192.168.0.82
Ground Floor	Reception (Guest WiFi)	2	VLAN 12	192.168.0.83 - 192.168.0.84
		Server Count		
Second Floor	Server Room	5	VLAN 101	192.168.0.65 - 192.168.0.69

Table 2 Network Device IP Range

Field	Details
IP Address	192.168.0.1 - 192.168.0.62
Network Address	192.168.0.0
Usable Host IP Range	192.168.0.1 - 192.168.0.62
Broadcast Address	192.168.0.63
Total Number of Hosts	64
Number of Usable Hosts	62
Subnet Mask	255.255.255.192

Table 3Server IP Range

Field	Details
IP Address	192.168.0.65 - 192.168.0.78
Network Address	192.168.0.64
Usable Host IP Range	192.168.0.65 - 192.168.0.78
Broadcast Address	192.168.0.79
Total Number of Hosts	16
Number of Usable Hosts	14
Subnet Mask	255.255.255.240

Table 4Access Point IP Range

Field	Details
IP Address	192.168.0.81 - 192.168.0.84
Network Address	192.168.0.80
Usable Host IP Range	192.168.0.81 - 192.168.0.84
Broadcast Address	192.168.0.84
Total Number of Hosts	4
Number of Usable Hosts	4
Subnet Mask	255.255.255.252

Table 5Workstation / PC IP Range

Field	Details
IP Address	192.168.0.129 - 192.168.1.254
Network Address	192.168.0.128
Usable Host IP Range	192.168.0.129 - 192.168.1.254
Broadcast Address	192.168.1.255
Total Number of Hosts	512
Number of Usable Hosts	510
Subnet Mask	255.255.254.0

3.4. Current Progress and Tools

The network design has been built and tested using a variety of tools and technologies to ensure operation and security.

- GNS3
    - GNS3 is the primary tool used for the design, simulation, and testing of the new network topology. It allows for complex network environments to be created using multiple routers, switches, and firewalls, producing realistic simulation of network behavior.

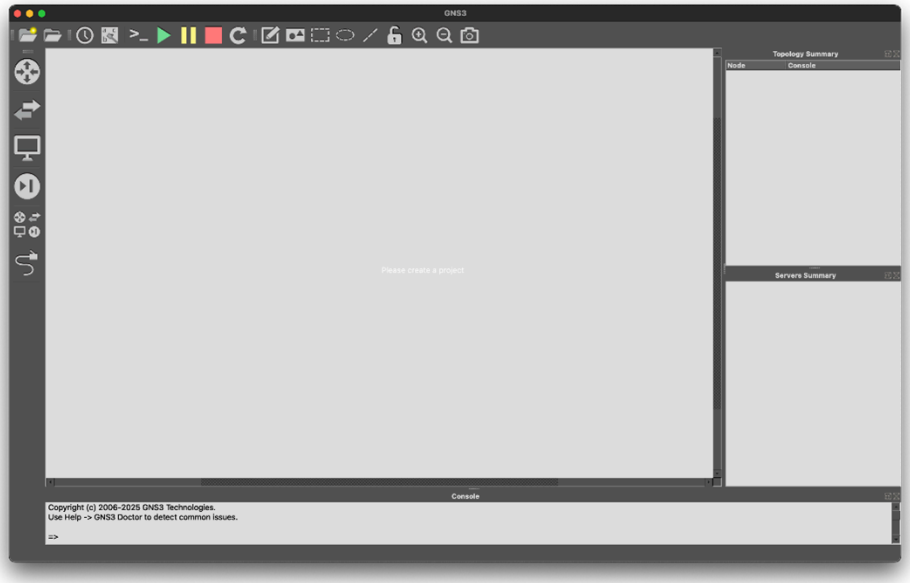


Figure 8 GNS3

- Cisco Packet Tracer
    - Used for testing smaller configurations and troubleshooting networks. It allows for testing of network devices in a simplified viewing, testing, and user-friendly interface.

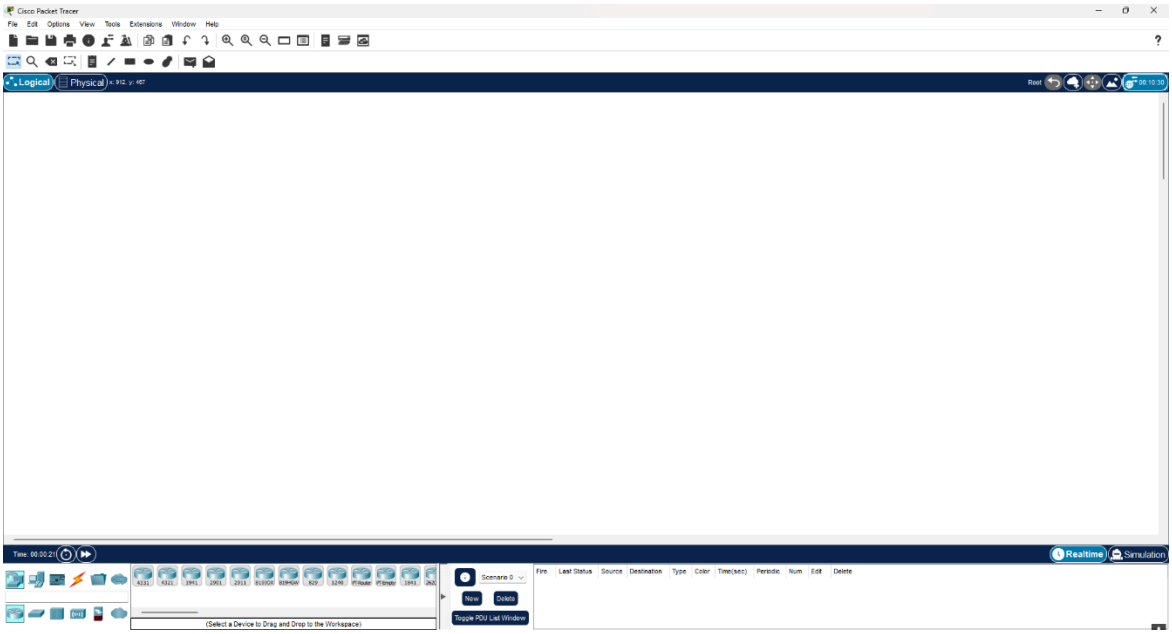


Figure 9 Cisco Packet Tracer

- VMware Fusion
    - Used to operate the GNS3 VM on a Mac to provide an optimized running and isolated environment for testing and simulation.

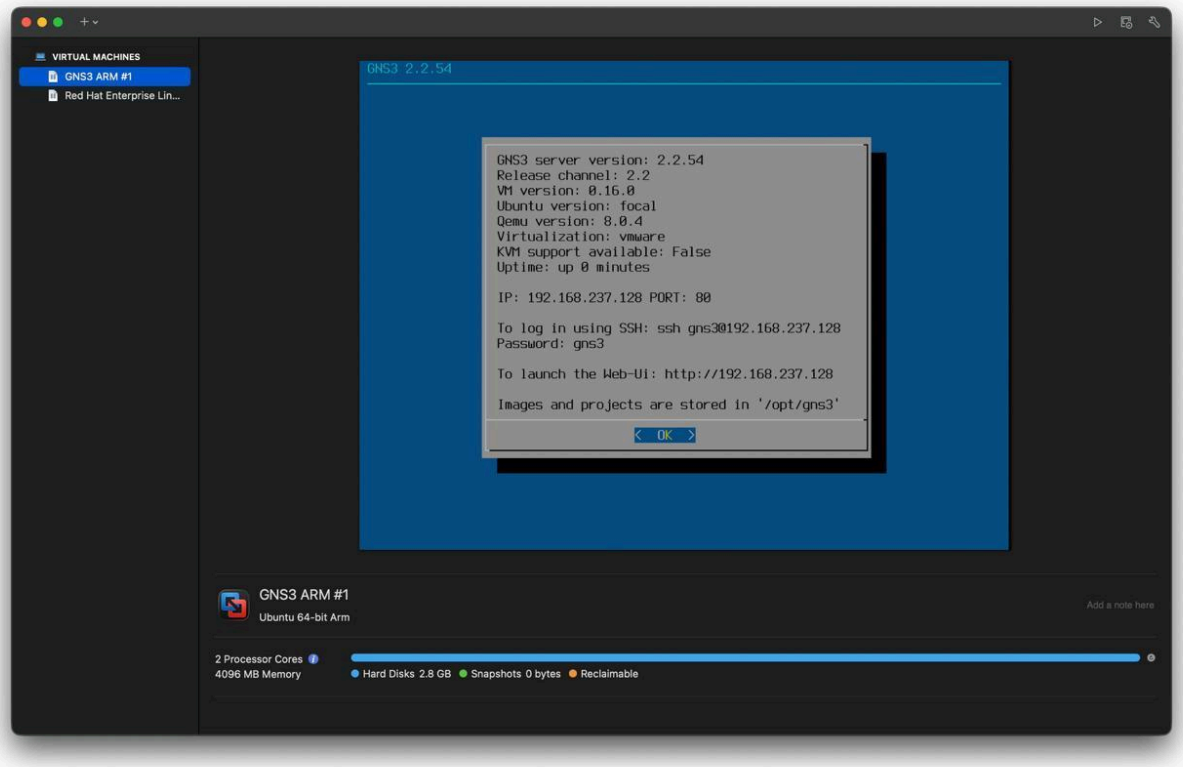


Figure 10 VM Ware Fution

- VMware Workstation Pro
    - On the Windows PC, VMware Workstation Pro is used to configure servers and run Windows Server 2022 to manage network services.

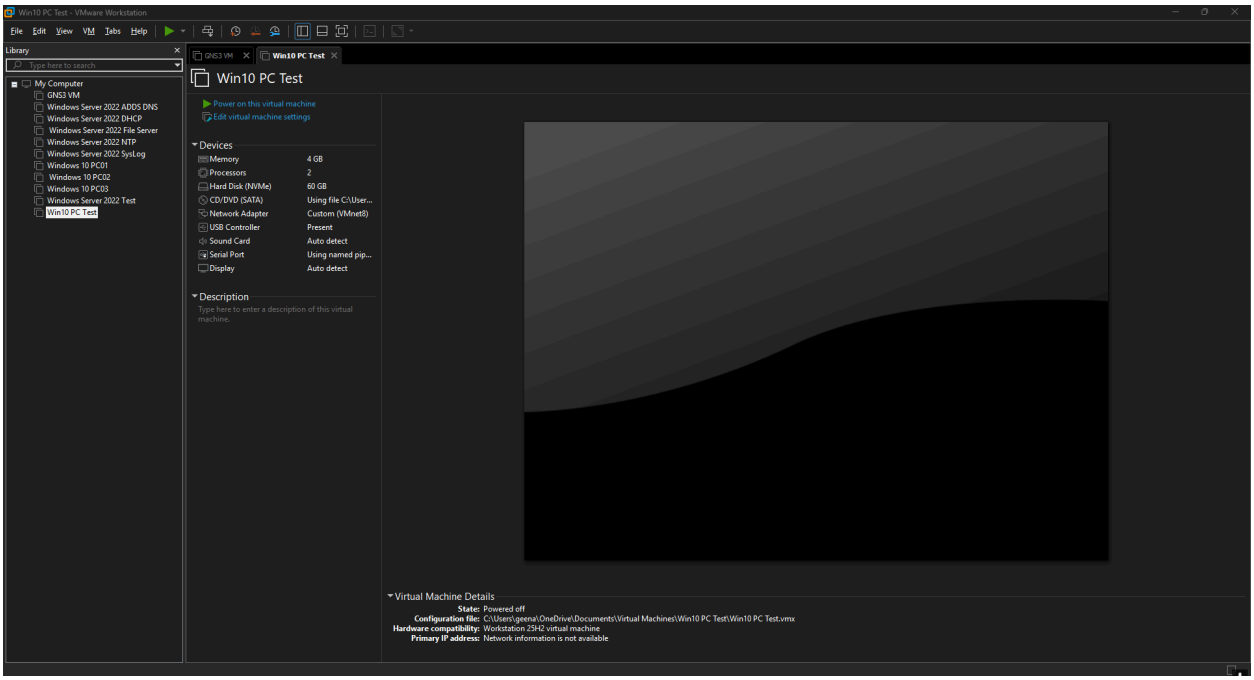


Figure 11VMware Workstation Pro

- Network Devices
    - The following are the network devices mainly used in GNS3 to simulate the network.
        - FortiGate Firewall: FortiGate provides advanced network security with capabilities such class deep packet inspection and threat intelligence integration.
        - Cisco c7200 Router: used to route traffic between VLANs and to ensure traffic is properly segmented in the network.
        - Cisco IOU L2 and L3: used to simulate L2 and L3 network switches to ensure the efficient management of traffic and provide security.
- Backup Systems
    - Redundant components are integrated into the design, including backup ISPs, firewalls, and switches, which are configured to automatically take over in the event of a failure.

4. Product Implementation

4.1. Implementation Overview

The redesign project for the NIC network for ABC Holdings PLC was performed in stages, starting with the configuration and set up of key network components and then moving onto testing and validating the entire design. The design included the segmentation of VLANs, firewalls, IDS/IPS, DNS, DHCP, FTP, and Syslog Servers. As these components all needed to work together, it was critical to perform this build and test in a contained simulated environment which was achieved using GNS3, Cisco packet tracer, and VMware Workstation. The network was tested in various scenarios to validate the operational functionality, security, and scalability.

4.2. Tools and Technologies Used

The implementation phase utilized the following you the following tools and technologies.

- GNS3
    - This was the primary tool used for the design and simulation of the network infrastructure. GNS3 made it possible to design and build virtualized environments including routers, switches, and firewalls.

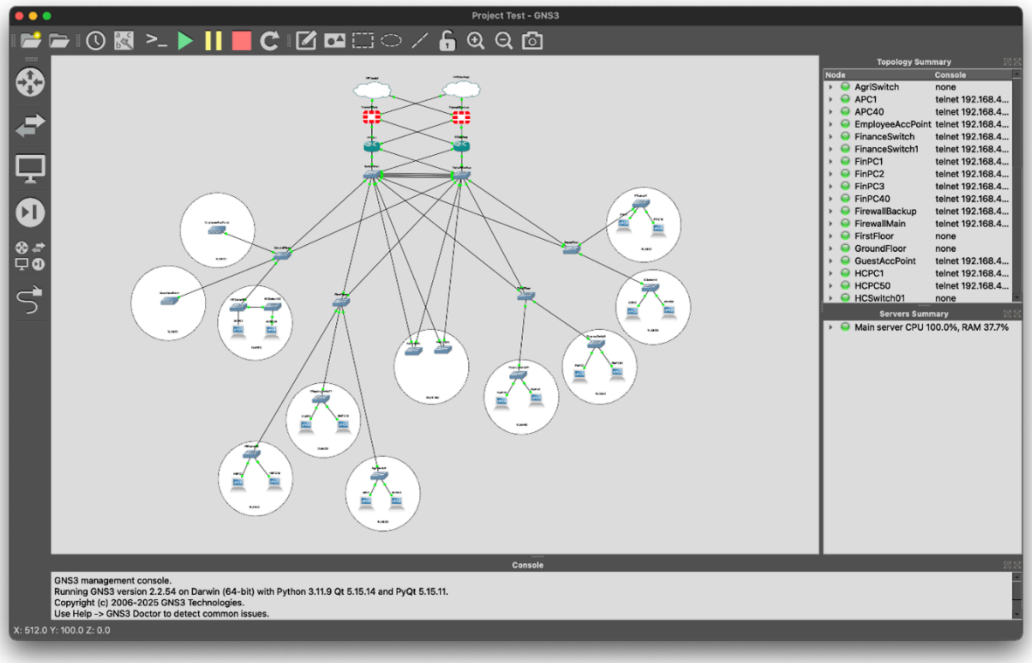


Figure 12GNS3 network topology

- Cisco Packet Tracer
    - Utilized when testing smaller configurations and troubleshooting configurations in the simulation.
- VMware Fusion
    - Run on the Mac, this tool allowed the operation of the GNS3 VM as well as to create a dedicated and optimized simulation environment.

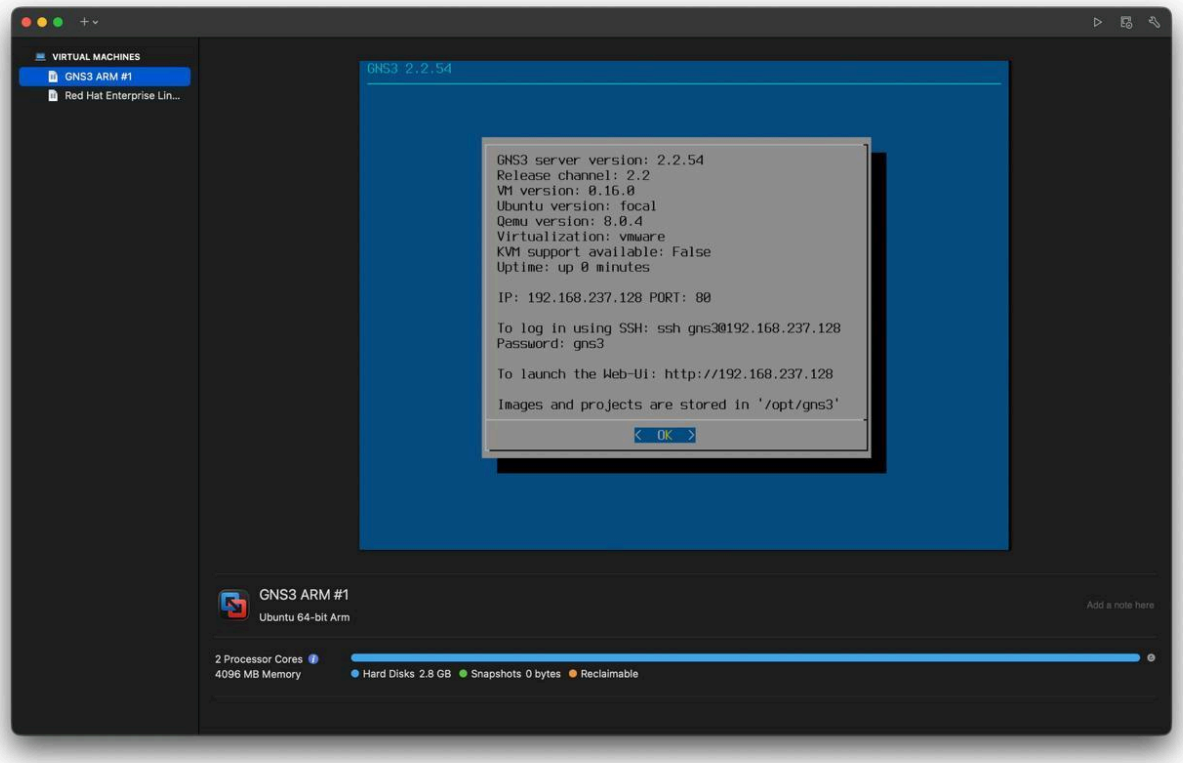


Figure 13VMware Fusion setup

- VMware Workstation Pro
    - On the Windows PC, VMware Workstation Pro was used to configure and manage servers, such as Windows Server 2022 for DNS, DHCP, FTP, and SysLog services.

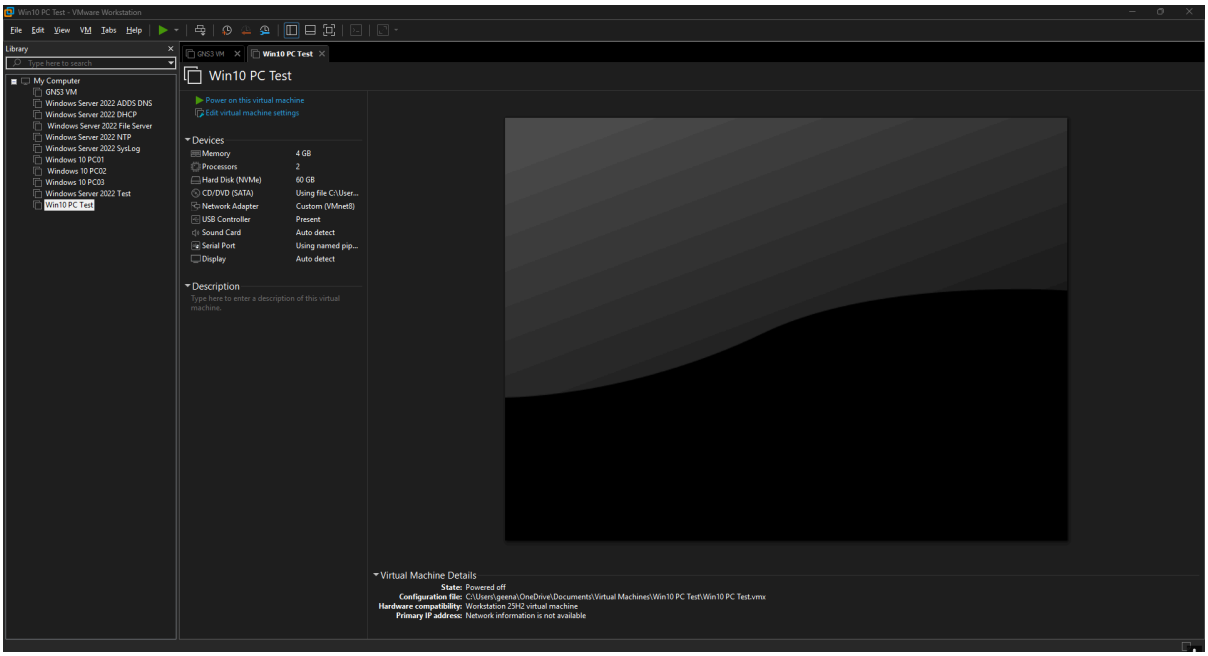


Figure 14VMware Workstation Pro configuration 01

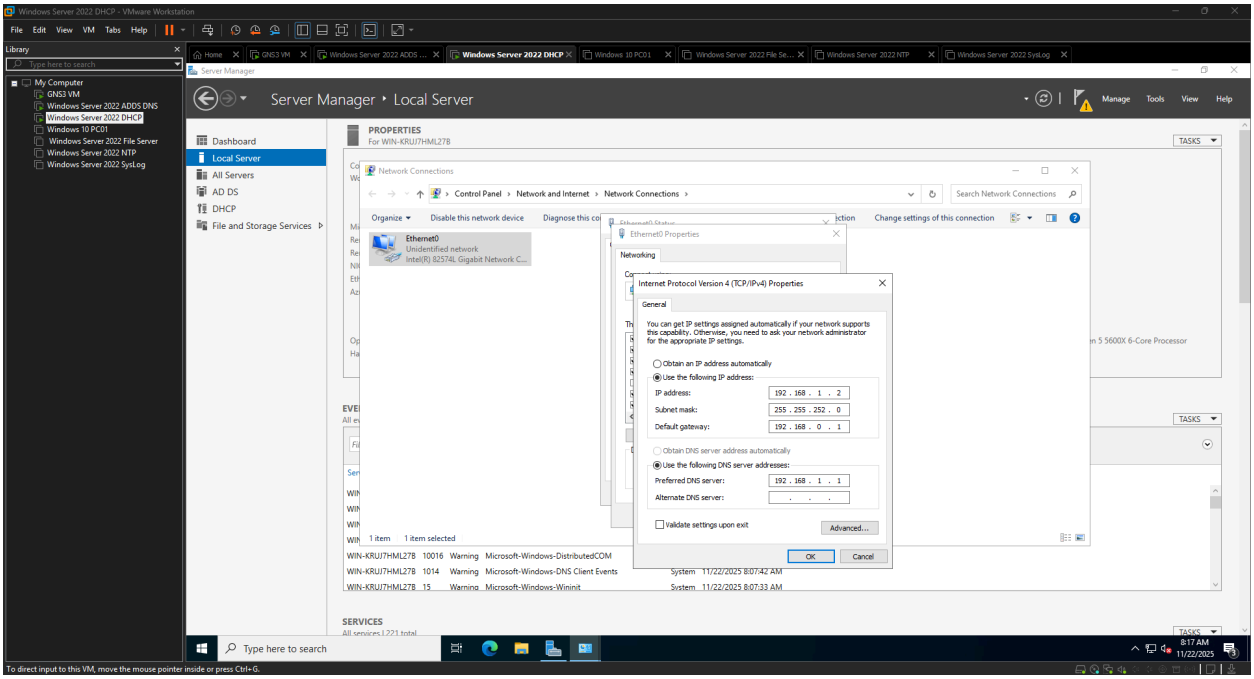


Figure 15VMware Workstation Pro configuration DHCP Server

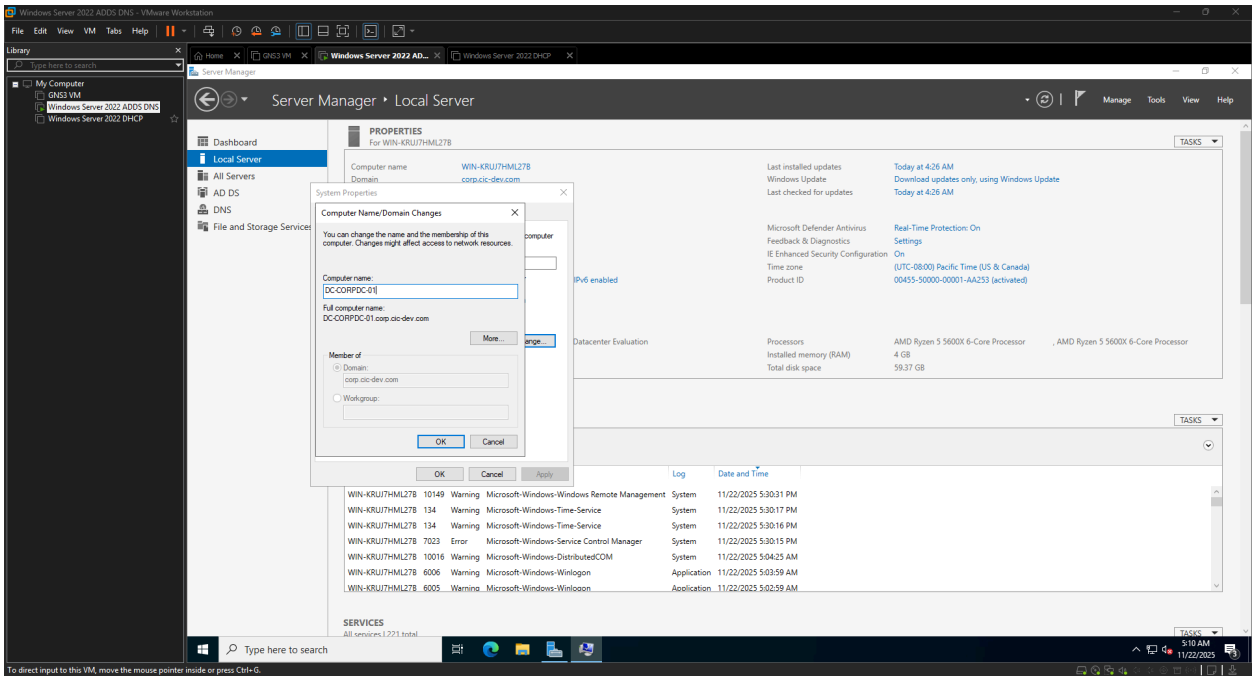


Figure 16VMware Workstation Pro configuration DNS Server

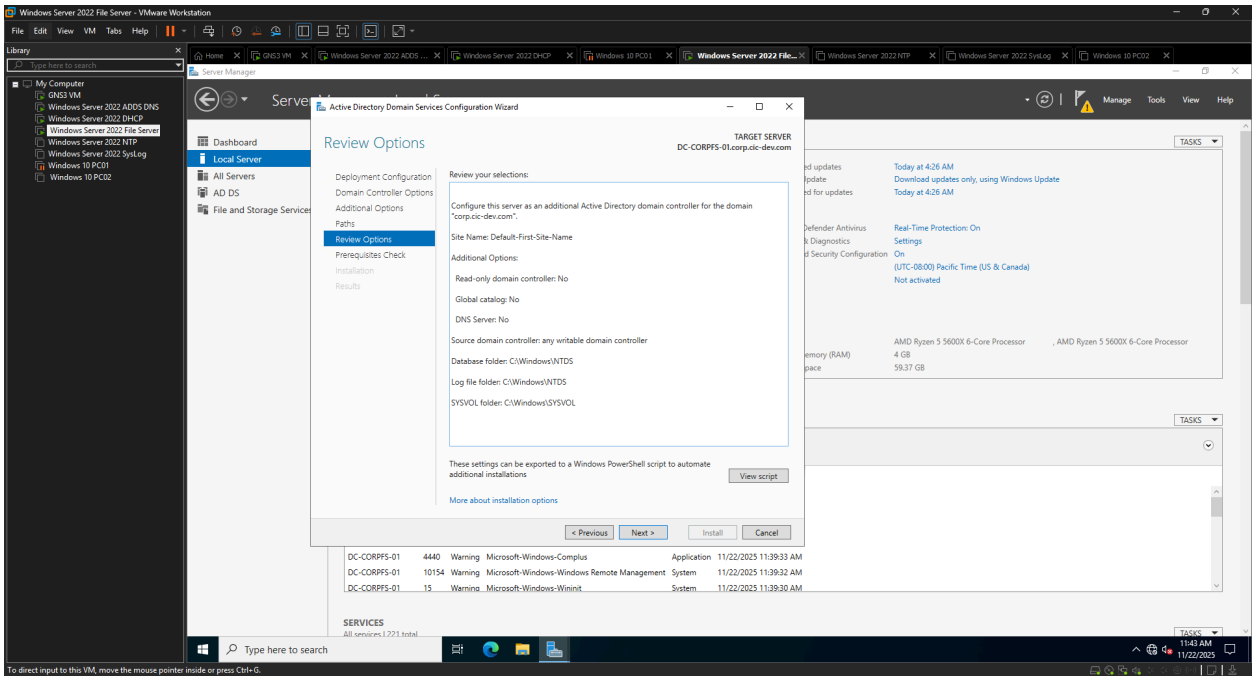


Figure 17VMware Workstation Pro configuration FTP Server

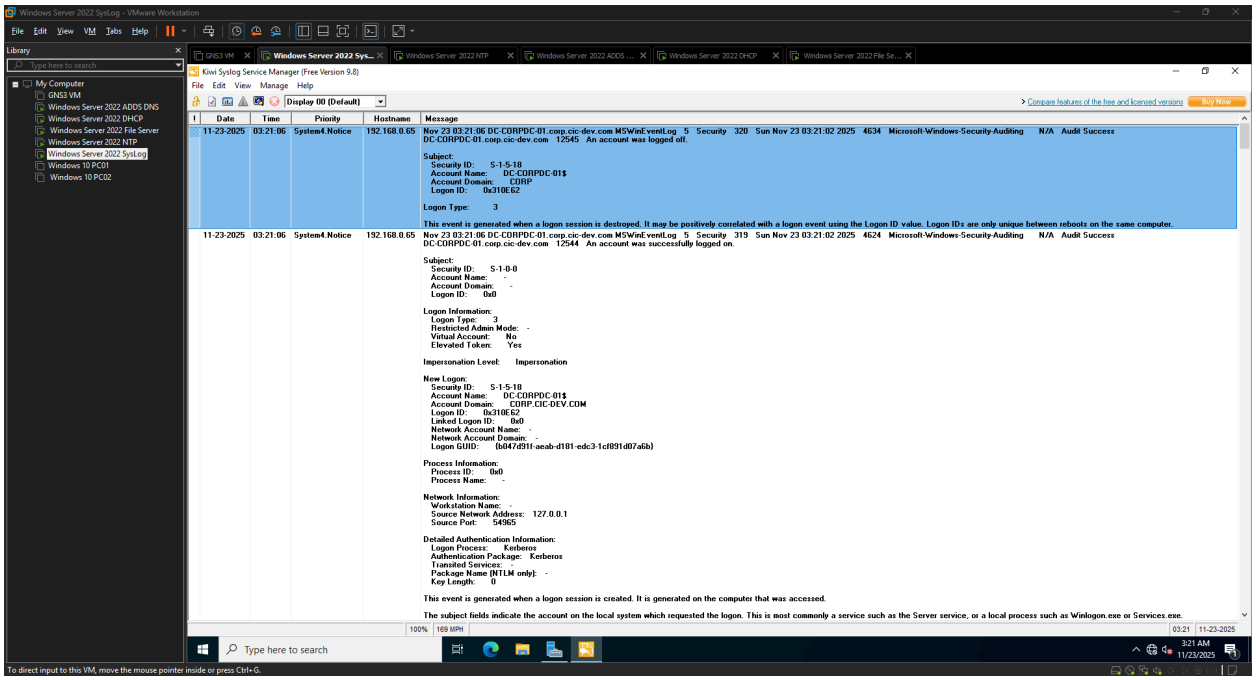


Figure 18VMware Workstation Pro configuration Syslog Server

- FortiGate Firewall
    - Deployed to manage traffic and provide security, including IDS/IPS functions.



Figure 19FortiGate firewall configuration 01

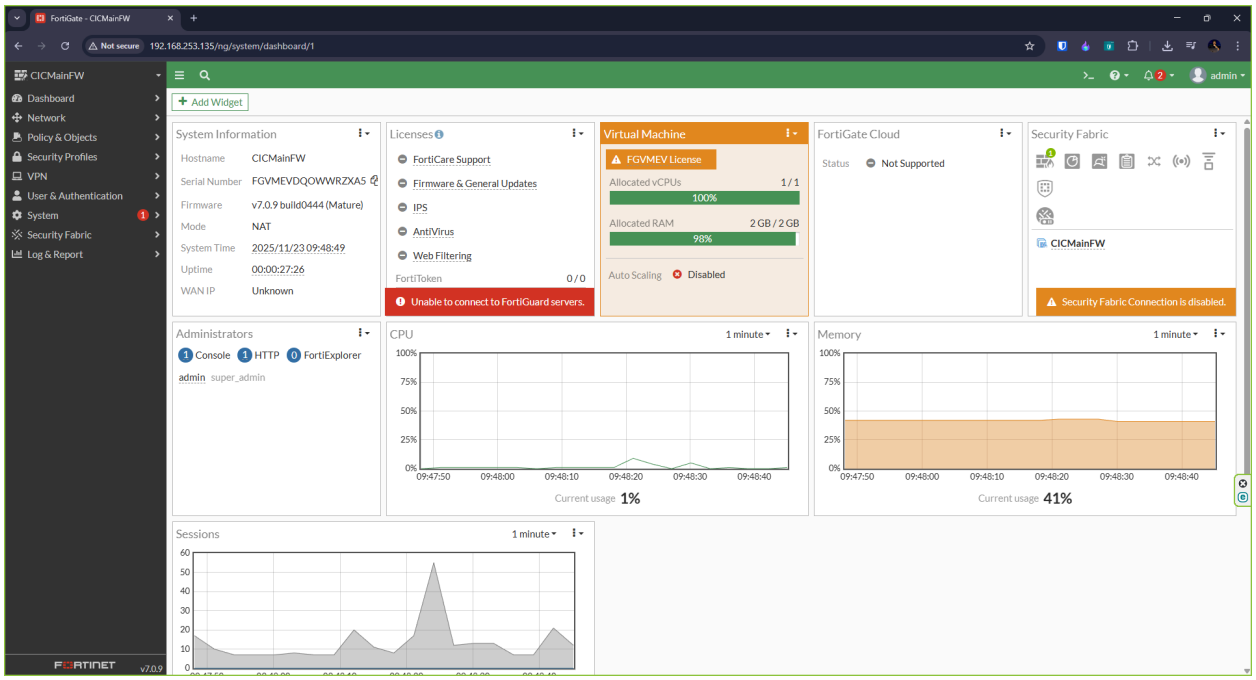


Figure 20FortiGate firewall configuration 02

5. Validation

5.1. Testing Strategy

While formal performance benchmarking or formal penetration testing was not conducted, the testing of the newly redesigned network was performed through testing and verification of configurations. For the VLAN Segmentation, each department had its own VLAN assigned to ensure sensitive information was protected by segmentation where communication amongst departments would be limited. The VLANs were completed in GNS3, and the testing was achieved by simulating traffic flowing from various VLANs. The firewall also blocked unauthorized communication when policies were tested to validate that the VLAN segmentation was working properly.

In relation to the Firewall Configuration, the FortiGate Firewall was established to block unauthorized communication allowed between VLANs, allowing only legitimate communication. Furthermore, the firewall was enabled with IDS/IPS features enabled to monitor and block malevolent traffic. The firewall was then tested with unauthorized access from both internal and external traffic to see if the policies were correctly implemented. The validity of the results confirmed that the firewall rules were working properly which monitored or blocked malicious traffic.

The Server Configuration, specifically DNS, DHCP, FTP, and SysLog, was validated. The DNS server was validated to ensure domain name resolution across the network and that networked devices could communicate with each other by hostname. The DHCP server was validated by checking that devices received IP addresses automatically without conflicting with other devices. The FTP server was validated for file transfer capability to ensure secure communication between departments. Lastly, the Kiwi Syslog Server was configured to capture and store logs from network devices and was validated to ensure logs captured and stored from the devices were available for security monitoring.

To verify the availability of the network configuration, Redundant Backup Systems for the ISP, firewall, and core switches were configured. Tests of the Failover scenario were performed to demonstrate the primary components had failed. The Backup Systems took over automatically without affecting multiple applications during the tests, demonstrating the redundancy configuration were proper.

5.2. Expected Outcomes

The anticipated results based on the design and configuration of the network were as follows,

- VLAN Segmentation
    - To maintain confidentiality and safeguard sensitive information from being accessed by any unauthorized personnel, dedicated VLANs (Virtual Local Area Networks) were utilized for each business unit. All Inter-VLAN communication was limited to pre-authorized communication only. The firewall has been setup to regulate and control traffic flows between the VLANs, allowing only traffic originating from a pre-authorized business unit and denying all traffic attempts to communicate across the VLANs without pre-approval. Firewall testing was also conducted to verify that unauthorized traffic attempts to communicate across the VLANs were denied as anticipated. This demonstrates that VLAN segmentation successfully protects sensitive information while maintaining the overall security of the company's network through degradation of the risk of exploitation by unauthorized personnel.
- Firewall Control
    - In order to ensure that only legitimate communication occurs between VLANs and to stop any unauthorized access, a firewall was used. The firewall has been programmed to allow traffic that has been authorized for departments or inter-department communication. Besides being a basic traffic controller, an Intrusion Detection/Prevention (IDS/IPS) system has been added to the firewall to detect and prevent malicious activities. The performance of the firewall was evaluated by simulating several methods of bypassing network security. Both internally and externally, the firewall was able to effectively block all access attempts to the network. Additionally, the IDS/IPS system correctly identified and prevented several types of malicious activity as planned.
- Server Proper Operation
    - DNS, DHCP, FTP, and Syslog servers were configured and validated through testing to confirm functional performance.
    - The DNS server was configured to serve as the primary means of resolving domain names to IP addresses, so that network devices can communicate with one another using DNS hostnames as opposed to using numeric IP address. Testing of the DNS server verified that all network devices could successfully resolve domain names to associated IP addresses and, correspondingly, could successfully use domain names to access resources hosted on the network.
    - The DHCP server is configured to automatically assign IP addresses to network devices. Testing of the DHCP server confirmed that all network devices received unique IP addresses assigned from the configured DHCP address pool, thus permitting all DHCP-enabled devices to connect and use the LAN successfully.

- The FTP server is configured to facilitate file sharing between departments. The FTP server was tested to ensure that files can be transferred securely and that users will only have access to files based on their department and associated VLAN.
- The Syslog server is configured to capture log information from all of the LAN devices (routers, switches and firewalls) that are part of the network. The Syslog server will capture all critical system events including login attempts and security alerts to enable centralized monitoring for network security issues. All servers were tested and validated for their specific purpose and are fully functional in creating a secure and stable network environment.

- Backup ISP
  - Backup Internet Service Provider (ISP), firewall, and switches established high availability and thereby promote continuous connectivity to network services. The backup ISP is set up to be automatically activated in a situation where the primary ISP has gone down. Failover testing for the backup ISP was performed through manually disconnecting the primary ISP from the network, and the backup ISP successfully activated without interrupting any of the business' network functions. Thus, businesses retain external connectivity despite an interruption of the primary ISP service.
- Port security
  - Port security has been configured on the Switches to allow only designated equipment access to the network. Each Port of the switch is limited to a given number of MAC addresses; only devices with an approved MAC address can connect to the Switch. If an unauthorized device attempts to connect to the network via a port, it will not be allowed onto the network, and if an unauthorized device has been plugged into the Port of the Switch, then the switch should prevent that device from being able to connect to the network. Test results were obtained after plugging several unauthorized devices into the ports of the switches, and the network blocked those devices from being able to connect to the network confirming that Port Security settings are effective.

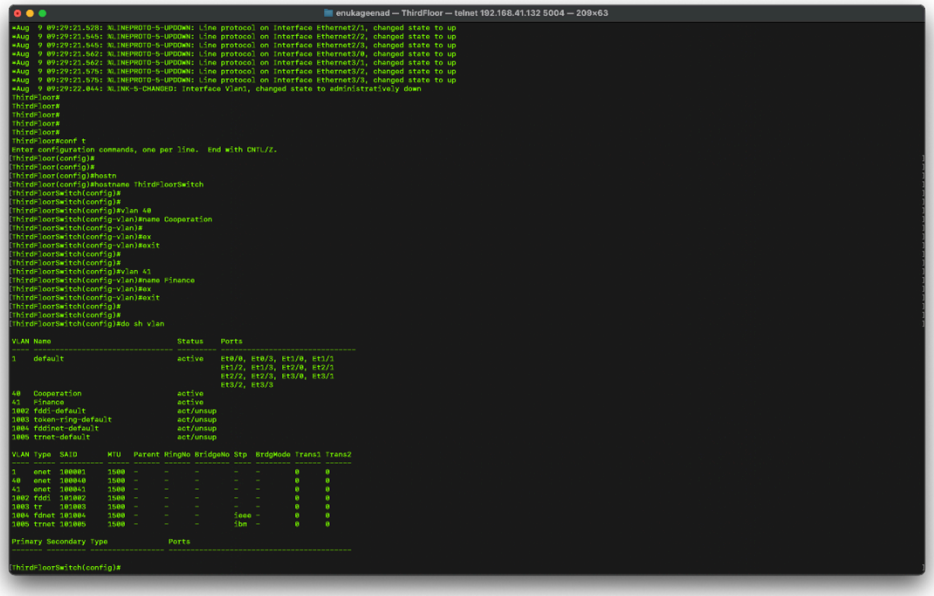


Figure 21VLAN Config01

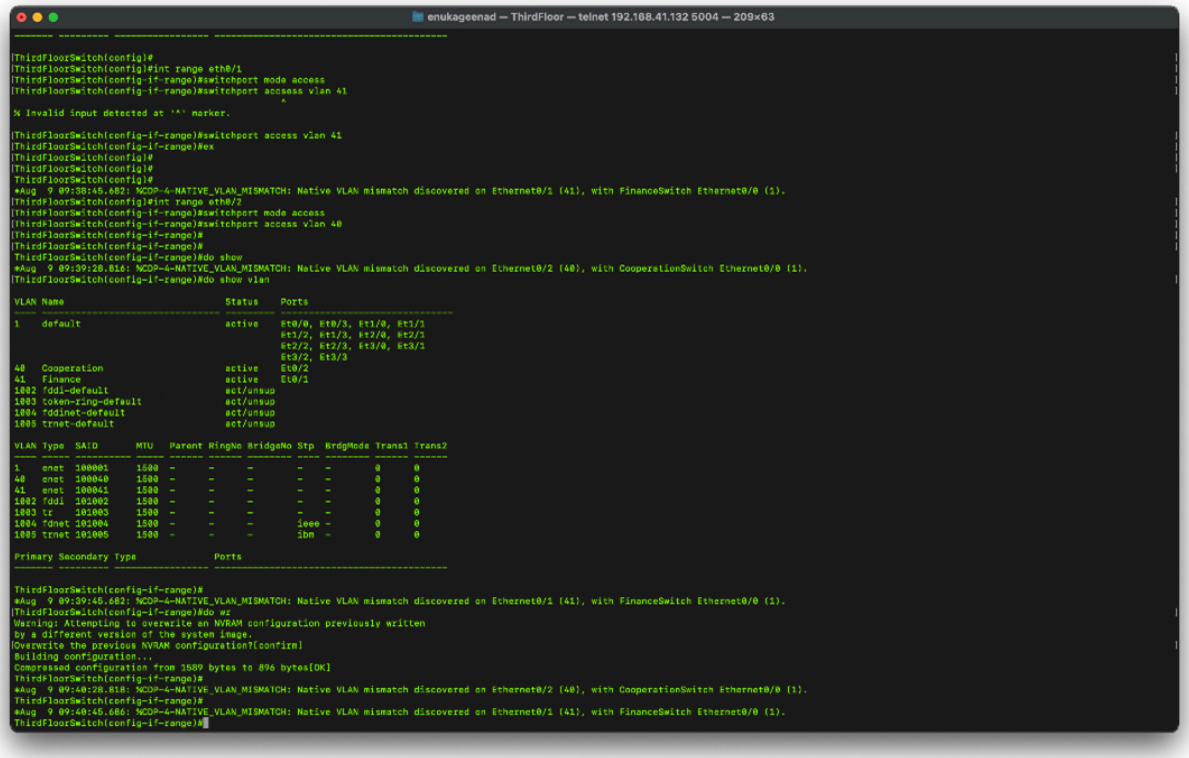


Figure 22VLAN Config 02

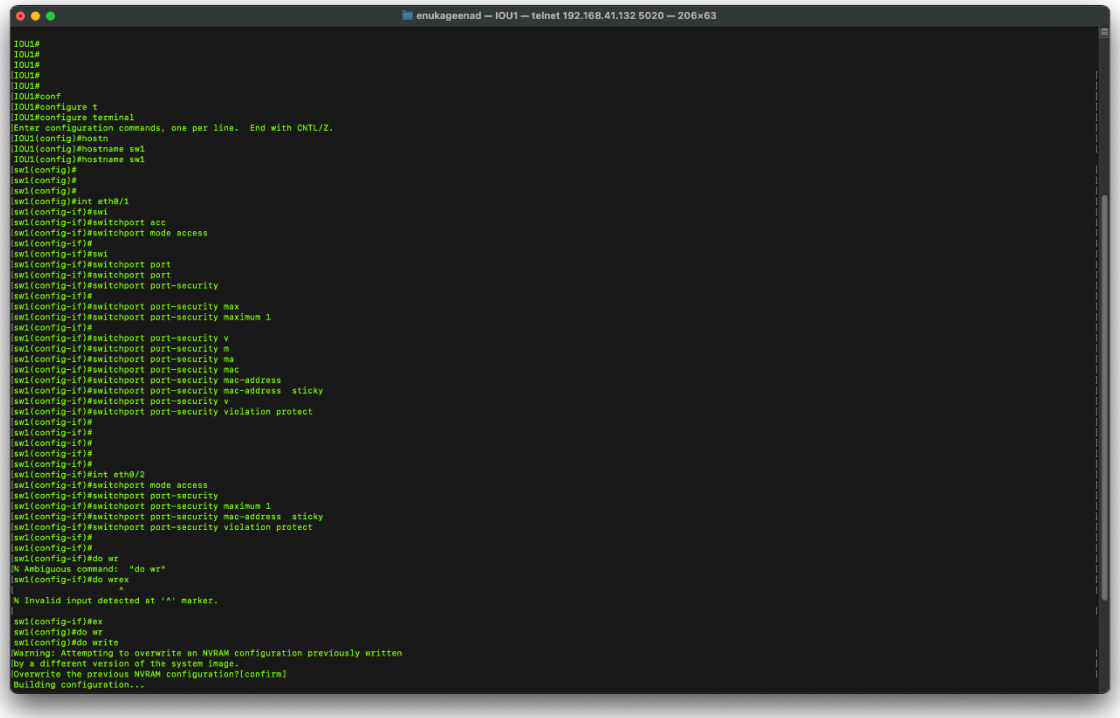


Figure 23Port security 01

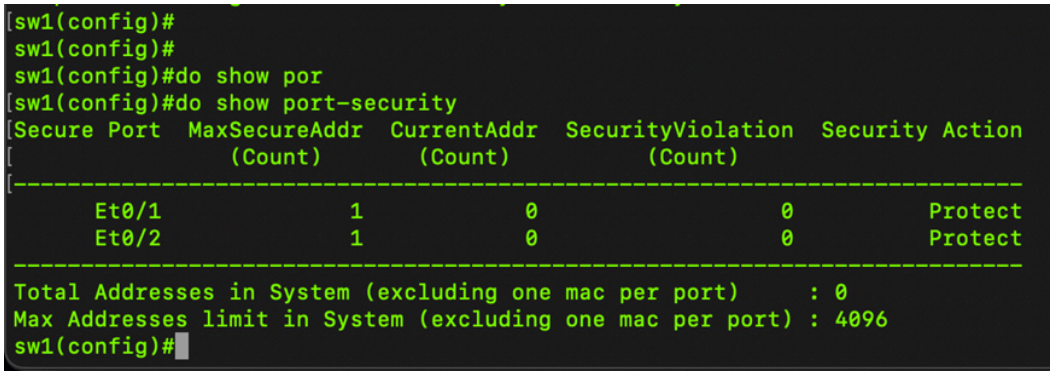


Figure 24Port Security conf 02

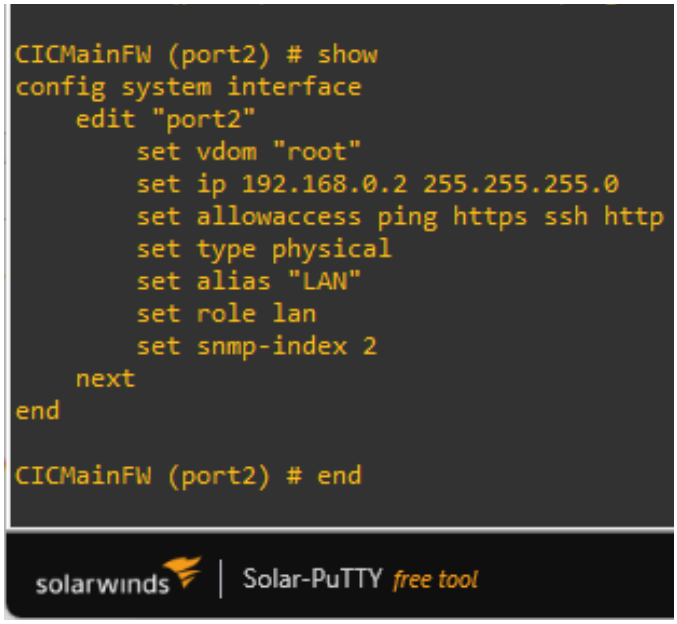


Figure 25Firewall

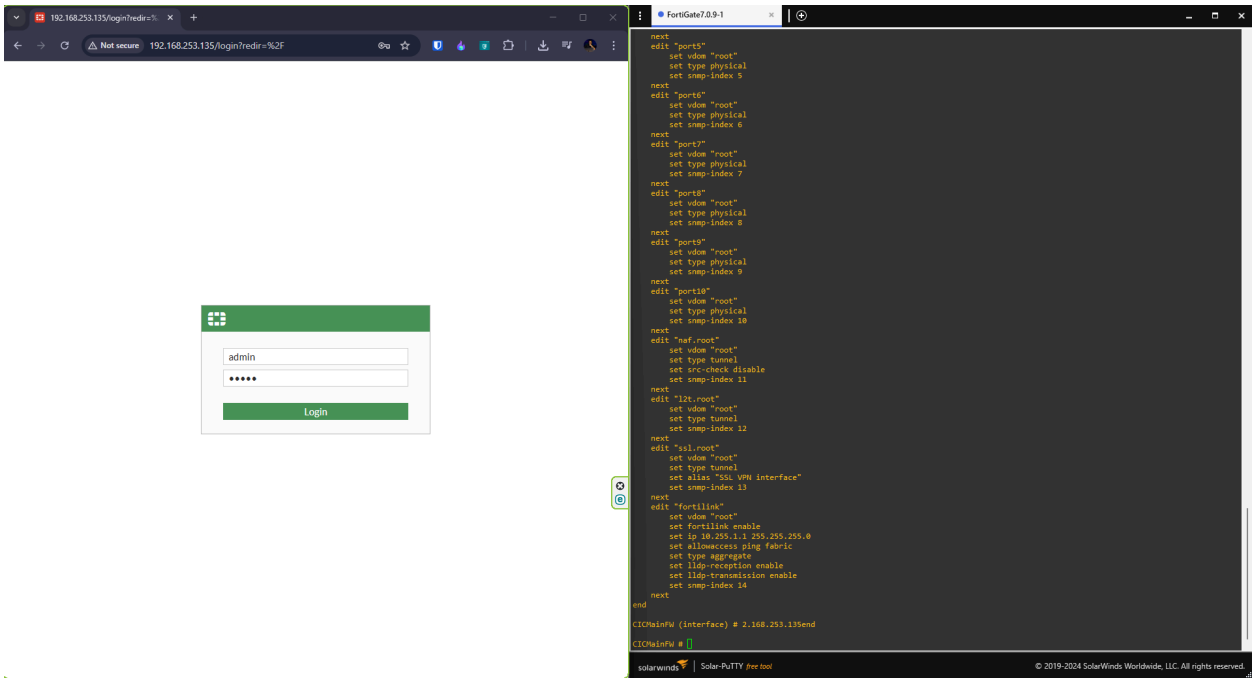


Figure 26Firewall CMD Config & Browser Login using IP

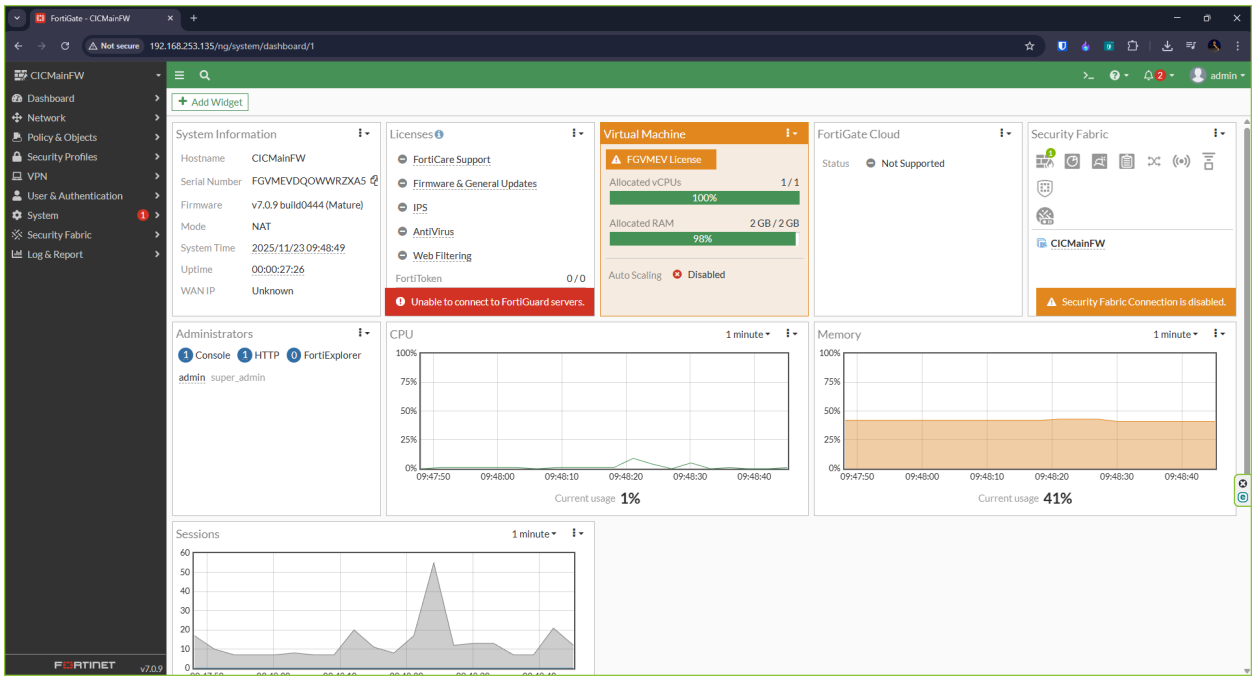


Figure 27Firewall Brouser Login Successful

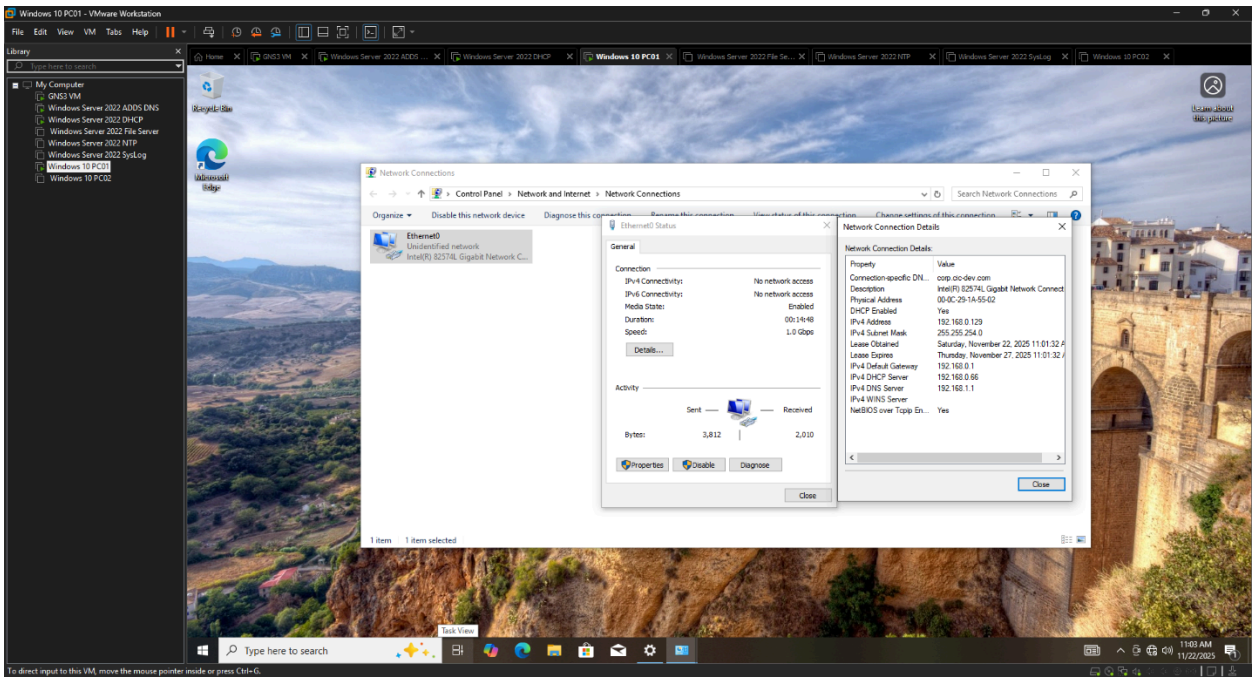


Figure 28DHCP Server Functioning in windows PC 01

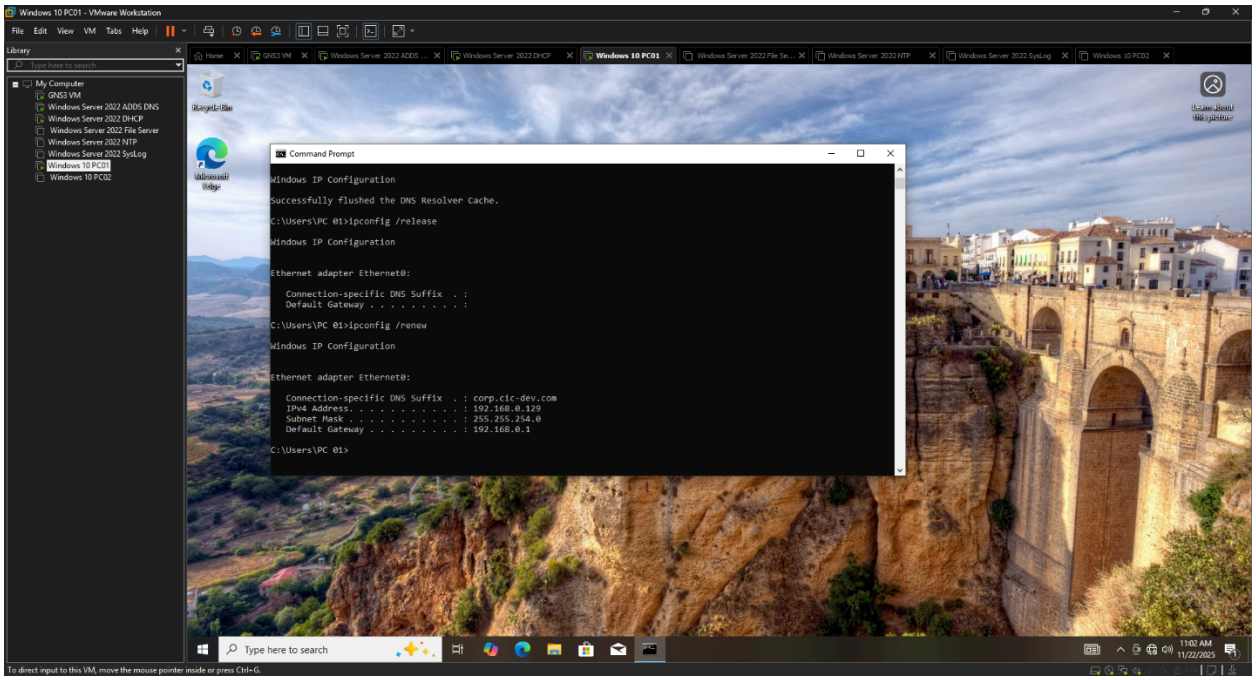


Figure 29showing DHCP Server Functioning in windows PC 01 using CMD

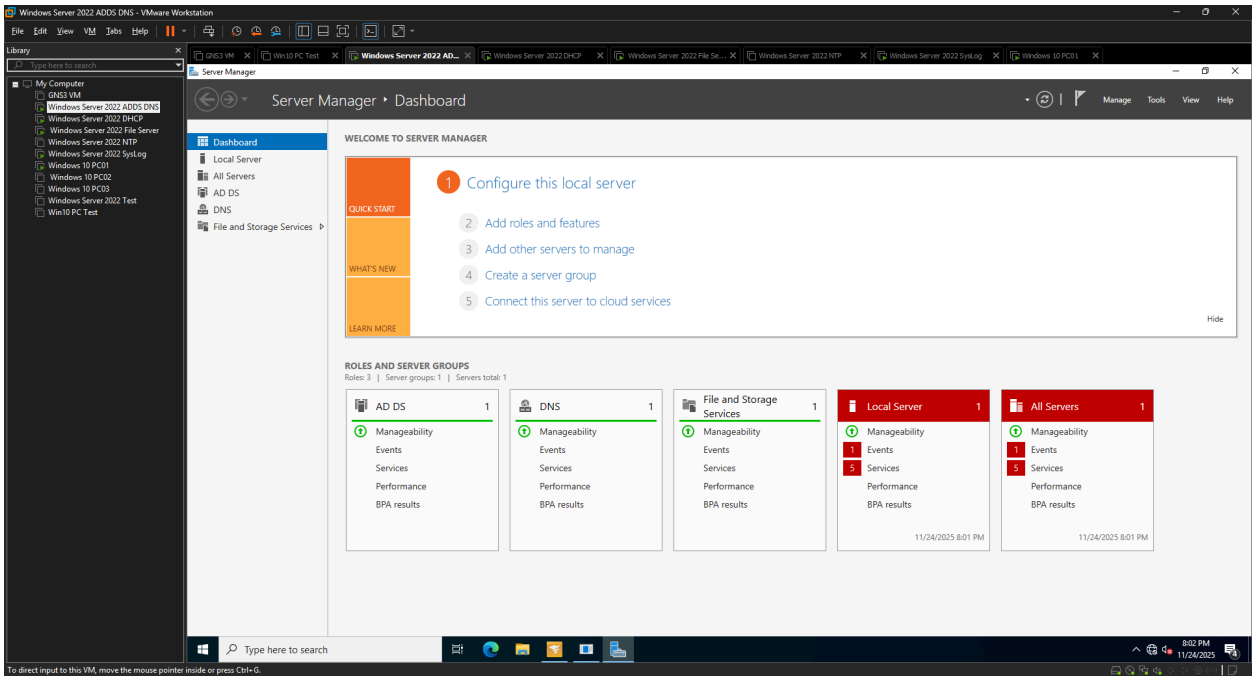


Figure 30DNS Server

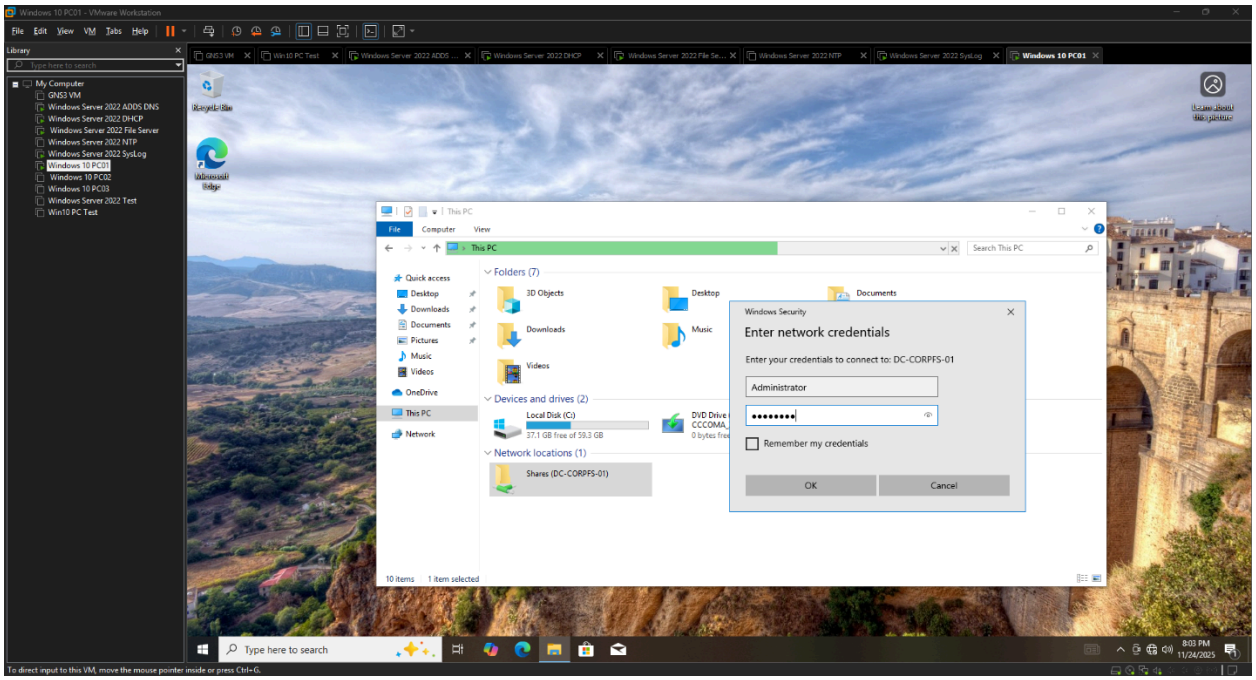


Figure 31FTP Server shared folder login on windows PC

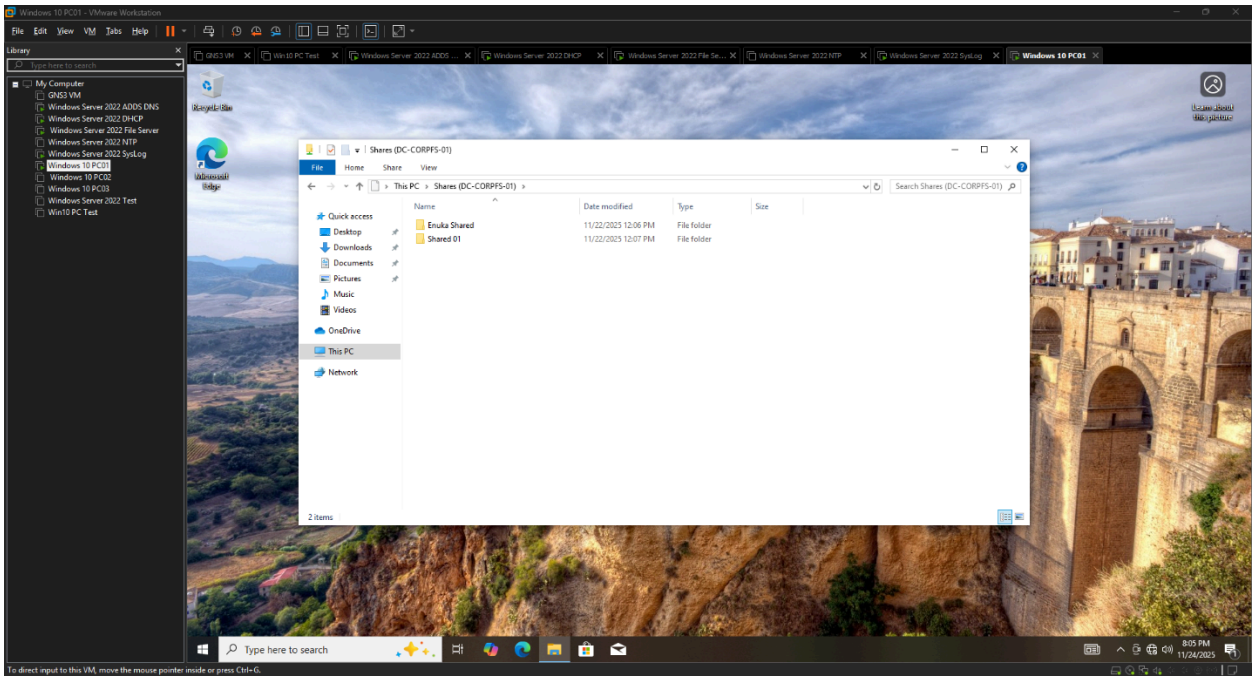


Figure 32 FTP Server shared folder shows on windows PC

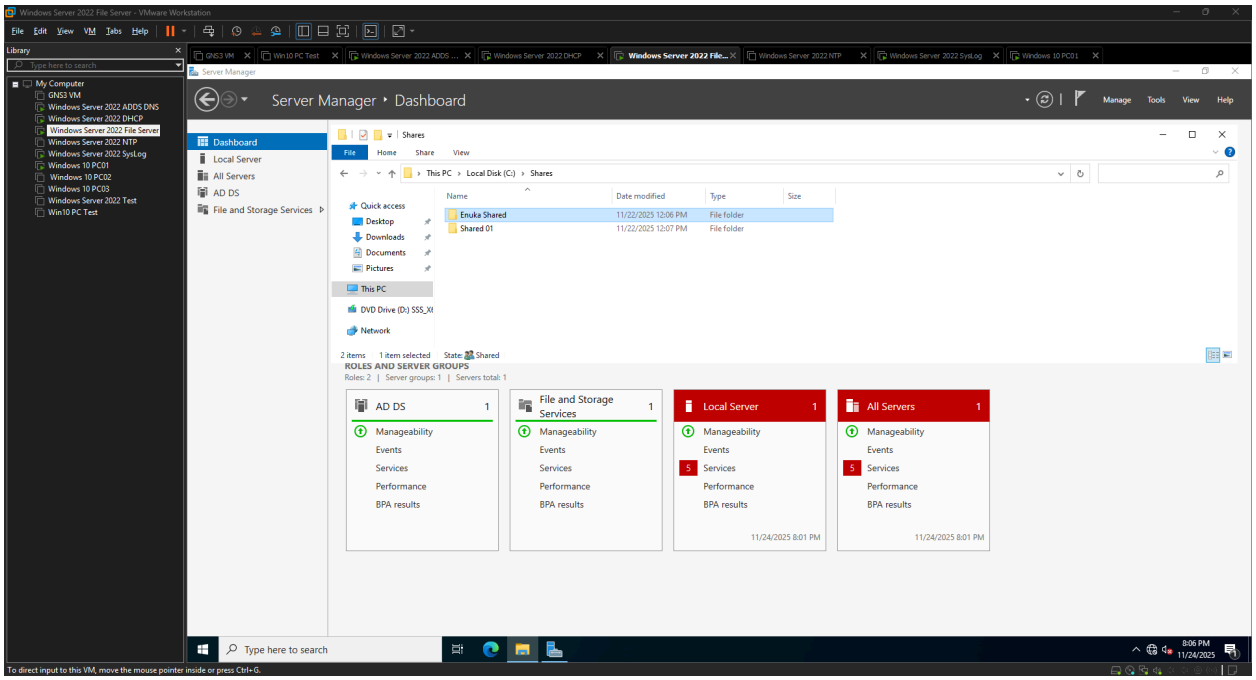


Figure 33FTP Server shared folder shows on Server

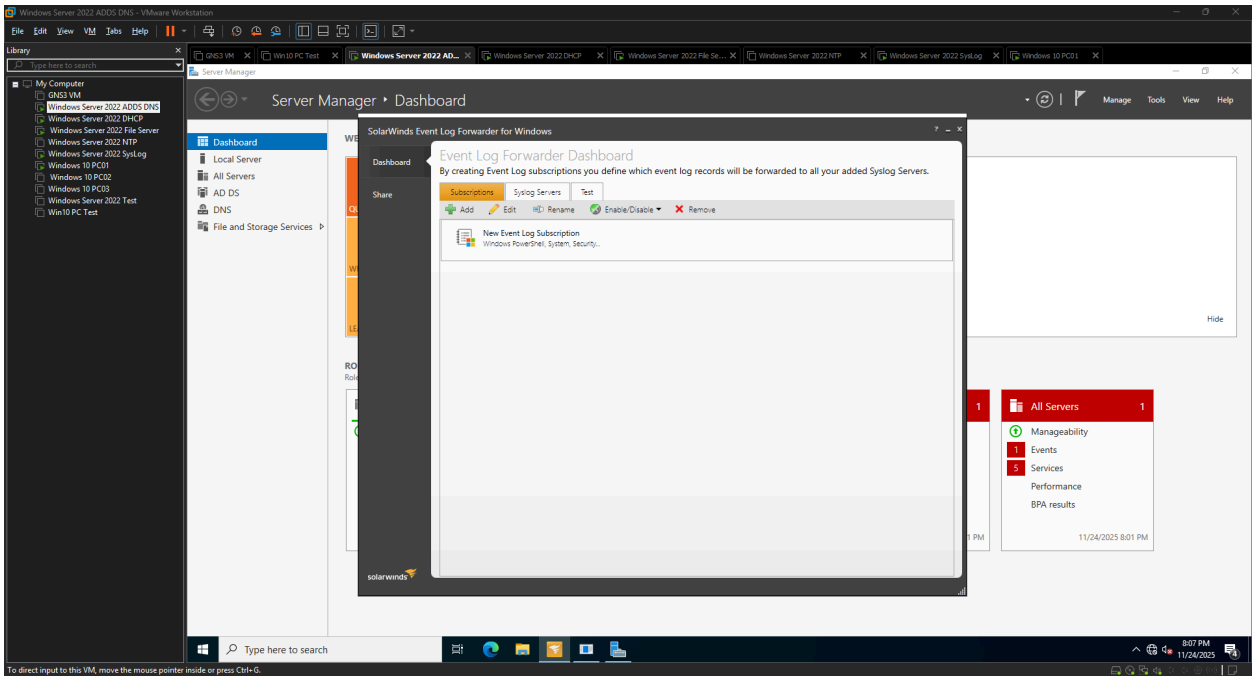


Figure 34 Setting up Syslog capture on DNS Server

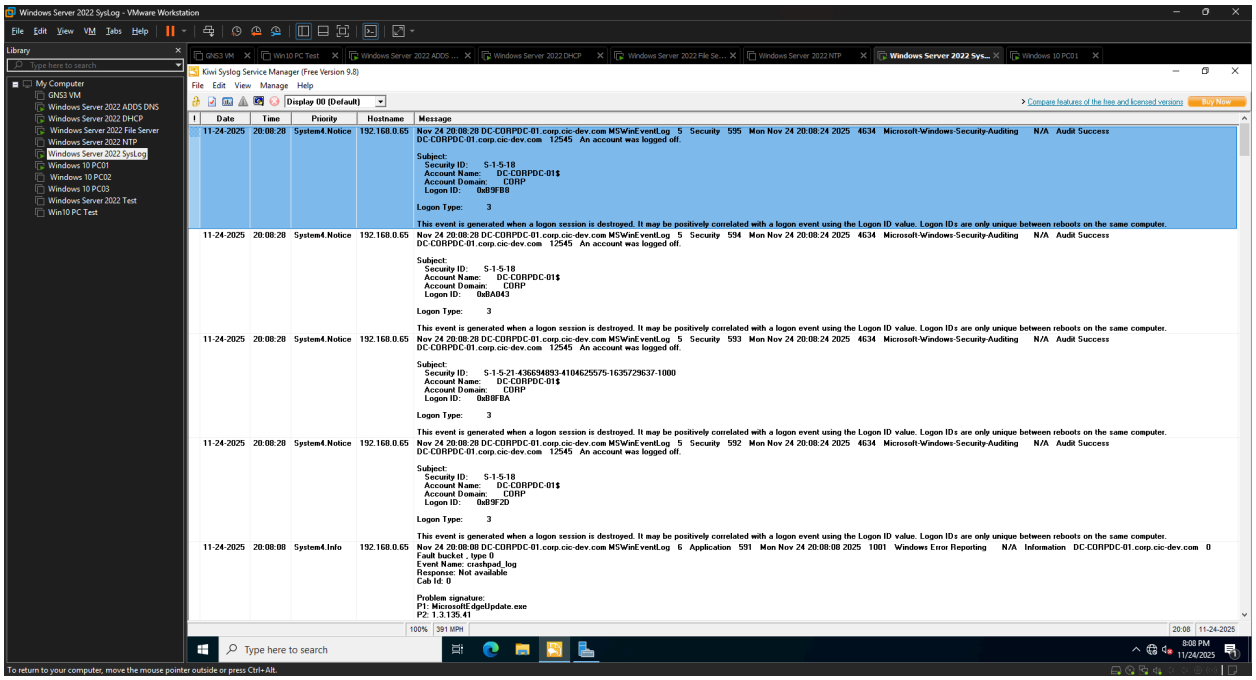


Figure 35Working Syslog server

5.3. Future Considerations

To promote the ongoing functionality and security of the network, the following actions need to take place:

- Regular Security Assessments
    - Security assessments must occur periodically to guarantee firewall rules, IDS/IPS configurations, and server settings are current and effective against newly emerging threats.
- Performance Monitoring
    - Using tools such as SolarWinds, real time network monitoring will immediately detect where performance bottlenecks occur and reach out proactively for troubleshooting.
- Scalability
    - The network is scalable enough to allow new VLANs, devices, and services to be added with relative ease, as the organization grows.
- Testing the Backup System
    - Backup systems need to be tested on an ongoing basis to ensure the failover is seamless and the network can remain available while a component fails.

6. 6. Critical Review & Conclusion

6.1. Closing executive summary

The goal of this project was to redesign the network infrastructure for ABC Holdings PLC to resolve multiple significant weaknesses within its existing network design. As part of the redesign, VLAN segmentation, firewall setup, IDS/IPS integration, and backup systems were developed to ensure the network was designed to be secure and scalable for the future.

- VLAN segmentation of departments into separate VLANs to improve data security and reduce the risk of unauthorized access.
- Firewall and IDS/IPS settings configured to protect against both outside and insider attacks with specific rules to deny unauthorized traffic.
- Backup systems were developed for the ISP, firewall, and switches to maintain network availability in the event of failure.
- Setup of the necessary services, including DNS, DHCP, FTP, and SysLog Servers with functionality and security verification tests.
- The network was tested using simulated traffic, security, and failover testing to confirm that the design met the business and security requirements.

Despite significant advancements, we encountered some challenges related to not only device compatibility, but also difficulty configuring the server running on the Mac and configuring VLANs and firewalls. This included: troubleshooting and configurations changes, as well as switching from our Mac to a Windows PC to manage server access. At the end of the day, we were able to overcome these challenges and keep the project on track to reach our design objectives.

6.2. Conclusion

ABC Holdings PLC’s redesigned network architecture is now sturdier, more secure, and scalable to meet the operational needs of the organization and future growth. The project worked to mitigate and address the main vulnerabilities that were discovered in the original network assessment which includes: lack of VLAN segmentation; ineffective Wi-Fi security; and lack of backup. The security of the network is definitively improved, and the design is compliant with current industry standards and regulations.

- The significance of having proper VLAN segmentation and firewall configuration to improve security and reduce attack surface based on poorly defined network segmentation.
- Ongoing and consistent monitoring of network devices and regular updates to maintain security and performance while improving protection against new threats as new technologies emerge.
- Benefits of adding backup systems to improve uptime and reliability by reducing downtime in the event of a failure.

Going forward, the network will need continuous maintenance and monitoring to sustain security against ever-changing cyber threats in the future. Future plans include the use of new technologies like AI/ML-based intrusion detection, scalability to enable cloud services, and regular security audits to validate compliance with universal security frameworks.

The work completed to this point has established a solid basis for ABC Holdings PLC's network, that will be well-equipped to address the future challenges of contemporary cybersecurity threats while fulfilling the requirements of the company in terms of expansion and operational requirements.

To sum up, this project has provided an opportunity for improvements to ABC Holdings PLC's network security and educational experiences. Future improvements may include the adoption of Zero Trust Architecture (ZTA), AI-based threat detection, and ongoing optimization of network performance. Continuous staff training and easily implemented protocols for network maintenance will be keyed to the network's resilience to be also and the long-term viability of the network.

7. References / Bibliography

Tharun Alvitigala and Madusanka Mithrananda (2024). Cybersecurity Awareness and Challenges in Sri Lankan Small and Medium-Sized Enterprises: A Comprehensive Survey. 1st International Conference on Advanced Computing Technologies (ICACT 2024). [online] Available at: *https://www.researchgate.net/publication/385281665_Cybersecurity_Awareness_and_Challenges_in_Sri_Lankan_Small_and_Medium-Sized_Enterprises_A_Comprehensive_Survey*.

Livera, L. (2025). Sri Lanka's cybersecurity and information security market is undergoing significant transformation. An analysis by Lahiru Livera - Cybersecurity Advisor. [online] Linkedin.com. Available at: *https://www.linkedin.com/pulse/cybersecurity-information-security-market-sri-lanka-strategic-livera-9lo5c*.

Security, C.C. for C. (2023). Top 10 IT security actions: No. 5 segment and separate information – ITSM.10.092. [online] Canadian Centre for Cyber Security. Available at: *https://www.cyber.gc.ca/en/guidance/top-10-security-actions-no-5-segment-and-separate-information-itsm10092*.

Admass, W.S., Munaye, Y.Y. and Diro, A.A. (2024). Cyber security: State of the art, Challenges and Future Directions. Cyber Security and Applications, 2(2). doi:https://doi.org/10.1016/j.csa.2023.100031.

Lavrov, E.A., Zolkin, A.L., Aygumov, T.G., Chistyakov, M.S. and Akhmetov, I.V. (2021). Analysis of information security issues in corporate computer networks. IOP Conference Series: Materials Science and Engineering, 1047(1), p.012117. doi:https://doi.org/10.1088/1757-899x/1047/1/012117.

Srikanth Bellamkonda (2021). Network Security Challenges in Remote Work Environments: Solutions for Protecting Data and Applications. International Journal of Advanced Research in Electrical Electronics and Instrumentation Engineering, 10(10). doi:https://doi.org/10.15662/ijareeie.2021.1010004.

Wang, P. (2022). Research on firewall technology and its application in computer network security strategy. Frontiers in Computing and Intelligent Systems, [online] 2(2), pp.42–46. doi:https://doi.org/10.54097/fcis.v2i2.3931.

Chiba, Z., Abghour, N., Moussaid, K., Lifandali, O. and Kinta, R. (2022). A Deep Study of Novel Intrusion Detection Systems and Intrusion Prevention Systems for Internet of Things Networks. Procedia Computer Science, 210, pp.94–103. doi:https://doi.org/10.1016/j.procs.2022.10.124.

Shaik, A. (2024). A Survey of Emerging Techniques for Large Networks of Virtual Local Area Networks (VLANs) with Benefits and Limitations. International Journal of Current Engineering and Technology, [online] 14(06). doi:https://doi.org/10.14741/ijcet/v.14.6.11.
Qureshi, S., Tunio, S., Akhtar, F., Wajahat, A., Nazir, A. and Ullah, F. (2021). Network Forensics: A Comprehensive Review of Tools and Techniques. International Journal of Advanced Computer Science and Applications, 12(5), pp.879–887. doi:https://doi.org/10.14569/ijacsa.2021.01205103.

Liu, C., Singhal, A. and Wijesekera, D. (n.d.). A LOGIC-BASED NETWORK FORENSICS MODEL FOR EVIDENCE ANALYSIS. [online] Available at: *https://csrc.nist.gov/CSRC/media/Projects/Measuring-Security-Risk-in-Enterprise-Networks/documents/logic_based_network_forensices_model-for_evidence_analysis.pdf*.
Giac.org. (2021). Available at: https://www.giac.org/paper/gsec/2568/securing-network-vlans-a-case-study/104420.